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**Hybrid Supply Chain Models in Swedish Grocery Retail: Balancing Cost Drivers,
Environmental Sustainability and Social Responsibility**

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Abstract

Background: The Swedish grocery retail sector transforms at an accelerating pace since customers seek change while regulators are establishing policies and digital innovation. The rise of sustainability as an important issue comes from the consumer value changes and Sweden's climate neutrality goals. The essential demand for sustainability now exists because hybrid distribution systems deliver emission reduction, cost minimization and improved inventory management. This study will explore the hybrid supply chains and the relationship between cost drivers, environmental sustainability and social responsibility.

Purpose: This thesis investigates whether and how the hybrid supply chain models impact cost minimization, environmental initiatives and social responsibility in the Swedish grocery retail sector. The thesis goal is to determine effective and new trade-offs in the hybrid supply chain that balance economic, environmental and social outcomes.

Method: The thesis uses a qualitative approach with a focus on a literature search. This search was conducted to explore the topic more and academic databases were used to explore the research topic. The empirical findings are based on seven semi-structured interviews with both floor workers and workers at an executive level.

Conclusion: Combining a centralized supply chain with decentralized distribution reduces emissions and drives cost savings, controlling retailers and alerting them to shifts in the market. Inventory strategies match product characteristics by allocating stock based methods to high volume goods and order based systems to items with limited shelf life. Environmental standards such as sustainable packaging, waste minimization and renewable energy use are integrated into the operations. The research confirms that following the Triple Bottom Line Framework (TBL) pillars, Economic success, Environmental sustainability, and social responsibility are connected and intertwined within a hybrid supply chain model.

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1. Introduction

The first chapter in the thesis introduced the topic by its background, problematization, research problem & purpose and important key definitions. These subchapters from chapter one are necessary to provide the reader with a full understanding of the topic and purpose of this paper.

1.1 Background

The Swedish grocery retail sector is currently encountering a transformative shift framed by growing consumer demands, regulatory pressures, and advancements in digital technologies (Rönnbäck, 2019). Hybrid chain models, which associate centralized logistics with decentralized operations, are positioned as an innovative approach to respond to this evolution. For instance, ICA Gruppen, Sweden's largest grocery retailer, recently combined a centralized logistics hub in Västerås, where they adjust decisions at individual stores (Svensson, 2021). It enables efficient inventory control while also allowing for local responsiveness, which is especially valuable when managing perishable goods or regional demand shifts (Safira, 2024). Likewise, Axfood's logistics subsidiary Dagab has created a fully automated central warehouse that uses AI-based forecasting in order to reduce waste, while also allowing stores to adapt their order locally (Axfood, 2021). These models are precious in balancing cost drivers and environmental sustainability, a broadening priority for businesses and consumers. Sweden's grocery retail market is dominated by major competitors such as ICA, Coop, and Axfood, holding approximately 90% of the market share (Rönnbäck, 2019). Regardless of the dominance of physical stores, e-commerce is becoming more successful, driven by increasing consumer preferences for convenience and digital solutions. Despite this, online grocery sales still account for a slight portion of the market, showcasing logistical and operational challenges particularly to this sector (Rönnbäck, 2019).

Sustainability has become a major focus in Swedish retail, pushed by environmentally conscious consumers and strict regulations like the EU Green Deal and Sweden's plan to reach net-zero emissions by 2045 (Swedish Institute, 2024). Hybrid supply chain models play a crucial role in pursuing sustainability objectives. They allow localized fulfillment to reduce transportation distances and emissions while using centralized logistics to optimize inventory and minimize waste (Delecta, 2023). Technological improvements establish the effectiveness of hybrid supply

chains. Tools, for example, such as real-time inventory tracking, AI-based demand forecasting, and route optimization, support retailers in managing costs while reducing environmental impacts. Innovations such as this improve efficiency, authorizing a more precise alignment of supply with demand decreasing overstock and debris (Dalal, 2024).

The performance of effective supply chains is based on responsiveness and cost-efficiency, which comes from the facilities to inventory, transportation, and final retailers. These elements form decisions regarding sourcing and pricing, which might result in more competitive pricing for consumers or improved profit margins for businesses facing financial pressures (Bridges, 2024). Supply chains should be desired to maintain responsive and cost-effective levels to minimize cost and increase performance (Chopra & Meindl, 2015).

1.2 Problematicization

Hybrid supply chain models require careful planning and decision making to achieve economic, social, and environmental sustainability. This is a primary challenge that retailers in Sweden face because the relationship between reducing costs and adapting to the strict sustainability regulations in Sweden is colliding (Swedish Institute, 2024). Technology and innovation, such as developed electric transportation, distribution centers, and software, often come with an extreme investment or a sacrifice for the environment. Although these hybrid models offer a simple and effective solution to the challenges, it is still hard to meet the wants and needs of today's consumers and determine the best practices. (McKinsey & Company, n.d.). It is yet unclear how they are experienced and managed in practice within the Swedish grocery retail sector. The existing literature barely explores how decision-makers balance cost pressure and sustainability demands in real world scenarios (Höglund, 2025). Also, hybrid supply chains must address the tension between centralized efficiency and decentralized flexibility. As Clarke-Sather mentions, *“Centralizing production facilities may reduce costs but perhaps not environmental impacts. Decentralizing a supply chain may reduce transportation environmental impacts but increase capital costs”*. Decentralized fulfillment, for instance, as local sourcing and micro-warehouses, have the ability to enhance sustainability but commonly results in higher operational costs (Clarke-Sather, 2023). The digital transformation required to strengthen hybrid supply chains adds another layer of complexity. Retailers clash with integrating digital tools that facilitate real-time inventory tracking, route optimization, and demand forecasting. These technologies are

crucial for ensuring efficiency in hybrid models but require skilled labor and seamless system integration, which add additional expense (Lundström, 2023). This study will be valuable as it will explore how hybrid models can be structured to promote both cost control, environmental performance and social responsibility. If these models are better comprehended and implemented, they could help retailers minimize food prices, strengthen profit margins and contribute to national climate goals, therefore supporting economic competitiveness but also societal well-being (Liu, 2022).

1.3 Purpose and Research Question

The purpose of this study is to explore if and how hybrid supply chain models have an impact on cost-efficiency, environmental sustainability and social impact in the Swedish grocery sector. Strategies are being explored, outlining the best practices and identifying the trade-offs in hybrid supply chain adaptation.

To explore this topic, the following research question will be used to explore the topic:

How do hybrid supply chain models influence cost drivers, environmental sustainability and social responsibility in the Swedish grocery retail sector?

1.4 Definitions

Supply Chain Management: Is the coordination and management of all activities regarding producing and delivering a product or service, from sourcing raw materials to delivering the final product to the customer (Fernando, 2024).

Centralized Supply Chain: A model in which decision-making, inventory management and distribution are controlled from a single central location (Wale, n.d).

Decentralized Supply Chain: A model where operations for instance as warehousing and fulfillment are spread across multiple regional locations (Wale, n.d).

Make-To-Stock (MTS): A production strategy where goods are manufactured in advance based on forecasted demand and stored as inventory (Customer Analytics, 2022).

Make-To-Order (MTO): A strategy where production starts only after a customer order is received (Customer Analytics, 2022).

Triple Bottom Line (TBL): A sustainability framework that evaluates business performance across three different dimensions: economic (profit), environmental (planet), and social (people) (Kenton, 2024).

Demand Forecasting: The process of predicting future customer demand for a product or a service, based on historical data, trends and market conditions (Salesforce, 2024).

Lead Time: The total time it takes for a product to move from the moment an order is placed until it is delivered to the customer (Kenton, 2024).

2. Literature Review

Chapter two provided the method of literature as well as all the concepts used in the paper. This laid the fundamental foundation of the literature research as well as the theories used to provide a strong, stable and fundamental foundation of the research topic.

2.1 Method of the Literature Review

The literature search for this study is managed by the use of Google Scholar, JU Primo, Scopus, and ResearchGate to ensure a comprehensive and credible collection of academic sources. These platforms were selected for their extensive coverage of peer-reviewed journal articles, books and conference papers applicable to the research thesis (Snyder, 2019). JU Primo, an academic database of Jönköping University, was for accessing scholarly journals, while Google Scholar offered a broad scope of academic sources, including open-access papers. Scopus was used to identify high quality research articles with convincing citation records, and ResearchGate was employed to discover relevant studies and working papers by experts within supply chain management and sustainability.

JU Primo's AI-powered search tools supported the initial identification of sources through generating a structured list of articles based on keyword applicability. While AI tools within JU Primo simplified the search process, they were purely used for retrieving available content and did not determine the assessment of the sources. A manual review was administered to identify the credibility of each article, considering factors such as the author's expertise, the publisher's reputation and the presence of peer review (Snyder, 2019). This mark was pivotal in mitigating the limitations of AI-generated results.

The search strategy was developed around selected keywords. The primary keywords involved "hybrid supply chain", "cost efficiency in supply chains", "supply chain sustainability" and "grocery retail supply chains". Further combinations, such as "hybrid supply chain AND sustainability " and "cost efficiency AND grocery retail supply chains" were used. To assure relevance to the Swedish market, "Sweden" was incorporated into keyword combinations to focus the review on local industry dynamics and regulatory conditions, as advised when contextual specificity is needed (Snyder, 2019).

Articles that lacked full-text access or did not address core themes for instance as cost drivers, hybrid supply chain models, environmental sustainability and social impact in retail contexts, Make-to-Order and Make-to-Stock strategies or centralized and decentralized structures were excluded. The most important objective was to identify articles that addressed these criterias and a special emphasis on Triple Bottom Line (TBL) framework, linking economic, environmental and social dimensions for the framing of this study.

The quality of the literature was evaluated based on journal ranking and relevance to the research topic. Preference was given to articles published between 1995 to 2025 and presented in high-ranking journals indexed in Scopus and ScienceDirect. Sources with significant citation counts were prioritized, as they illustrate great contributions to the field (Snyder, 2019). By following a systematic approach, the literature review ensures that the selected sources maintain credible, relevant and methodologically sound, constructing a comprehensive foundation for the study (Snyder, 2019).

2.2 Supply Chain Management

Supply Chain Management (SCM) demonstrates the coordination and combination of all activities involved in the production and delivery of goods and services, from raw material sourcing to final customer delivery (Fernando, 2024). It incorporates functions for instance as procurement, manufacturing, logistics, inventory management and the flow of information and finances beyond organizational boundaries. The goal of SCM is to develop value by improving efficiency, minimizing costs and enhancing customer satisfaction across the whole supply chain (Fernando, 2024). Naslund (2010) describes SCM as the management of a network of relationships within and between firms that expedite the forward and reverse flows of goods, services, information and finances to supplement value and improve customer satisfaction (Naslund, 2010).

2.2.1 Inconsistencies and Limitations in SCM and Practice

While SCM has become an important strategic tool for competitiveness in global markets, it is also a concept with various important challenges. As noted by Naslund (2010), there is no universally agreed-upon definition of SCM. It may lead to confusion among researchers and practitioners and makes it difficult to authorize common performance measures or best practices.

Some use SCM as a synonym for logistics or purchasing, while others describe it more broadly as the management of cross-functional and inter-organizational processes (Naslund, 2010). Using many different terms makes it hard to understand the topic. Terms for instance as “collaboration”, “integration” and “supply chain network” are frequently used but barely defined consistently. For example, collaboration can mean anything from basic information sharing to deep strategic partnerships involving shared risk and profit. This inconsistency makes it problematic to compare studies or replicate strategies across industries (Naslund, 2010).

Many SCM frameworks lay focus on one-to-one relationships (buyer-supplier), rather than on supply chain as a whole. Its narrow emphasis can lead to reduced overall performance and ignore the broader network dynamics that are increasingly important in global supply chains (Naslund, 2010). Additionally, a lot of empirical studies focus on specific industries such as automotive, limiting the relevancy of findings to other sectors such as grocery retail, where supply chains tend to be more disconnected. (Naslund, 2010). The integration of sustainability goals into SCM adds a layer of difficulty. Without any clear tools and metrics, it becomes problematic for firms to make strategic decisions that align with long-term sustainability objectives (Beske-Janssen, 2015).

Supply Chain Management is a wide and evolving concept that plays a central role in modern business strategy. As global supply chains continue to grow in size and importance, addressing these hindrances remains essential for both other researchers and practitioners (Sanchez-Flores, 2020).

2.3 Hybrid Supply Chain Models in Grocery Retail

The hybrid supply chain model has gained increasing attention in the grocery retail sector because companies pursue to balance cost efficiency with sustainability and adaptability (Hamzi, 2024). Hybrid supply chains implement centralized and decentralized frameworks, using both traditional and digital solutions in order to optimize inventory management, minimize environmental impact and improve customer service. Although the advantages, hybrid models present complexity and trade-offs that involve comprehensive analysis (Hamzi, 2024).

Hybrid supply chains arise as a response to evolving market demands, regulatory requirements and technological advancements. Normally, grocery retailers use either centralized or decentralized models. Centralized supply chains focus on efficiency by economies of scale,

reducing costs through bulk procurement and efficient delivery (Liu, 2024). Vice versa, decentralized models prioritize local responsiveness, enabling retailers to reduce transportation costs and enhance product freshness (Vallandingham, 2018). The hybrid model merges elements of both, optimizing cost structures while improving service degree (Liu, 2024)

The hybrid model seeks to leverage the best of both approaches, but it does not erase inefficiencies entirely. According to Lin (2019), hybrid supply chains emerge centralized and decentralized elements by implementing centralized data management platforms with decentralized operational facilities (Lin, 2019). Centralized features such as the use of organizational information systems and centralized control for inventory visibility and supply distribution (Lin, 2019). Decentralized elements involve distributed storage locations and local fulfillment units that allow for swift response to disruptions and customers needs. The hybrid structure allows for flexible changes, to allow centralized systems to monitor and plan while decentralized components execute locally with greater flexibility (Lin, 2019).

2.3.1 Balancing Efficiency, Sustainability and Technology Integration

Centralized logistics still constitute risks for instance as stockouts and demand variability, while decentralized fulfillment might lead to high operational costs and inefficiencies in scaling. Hence, the effectiveness of hybrid models relies largely on how well firms can implement their logistical and technological capabilities (Nakandala, 2019).

The implementation of hybrid supply chain strategies requires balancing efficiency and sustainability. Vallandingham (2018) highlights that grocery retailers must accommodate enabling technologies that are improving demand forecasting, inventory management and route optimization, by that means improving both cost efficiency and environmental sustainability (Vallandingham, 2018). Although, relying too heavily on technology poses new risks. High dependency on AI-driven forecasting may lead to inaccurate demand predictions, especially in volatile markets like fresh groceries (Hübner, 2013). A pivotal advantage of hybrid supply chains is their capability to align with sustainability goals. The flourishing consumer preference for low-carbon products and locally sourced goods demand a shift from purely cost-driven logistics to hybrid models that merge sustainability into supply chain planning (Liu, 2024). Zhou (2018) emphasizes that sustainable supplier selection has to consider economic and environmental factors but also social responsibility, embracing the need for hybrid approaches that combine

Triple Bottom Line (TBL) principles (Zhou, 2018). Although, sustainability efforts in hybrid supply chains tend to come at a financial premium. A great number of retailers struggle to balance the higher costs of sustainable products with the pressure to keep low prices (Nakandala, 2019).

2.3.2 Challenges and Gaps in Implementing Hybrid Supply Chains

While hybrid supply chains provide numerous benefits, it also poses new challenges in coordination, cost management and digital integration. Valladingham (2018) mentions that retailers have to handle the challenge of integrating new technologies while maintaining supply chain flexibility. The investment in digital tools and decentralized fulfillment centers may be capital-intensive, forcing companies to create long-term strategic plans (Valladingham, 2018). Retailers must providently navigate the trade-offs between centralized efficiency and decentralized flexibility. Deciding an optimal balance includes firms to consistently review their operational footprint, market demands and regulatory constraints. Simply integrating a hybrid model does not guarantee success (Davies, 2018). Firms have to be frequently proactive in adapting their strategies based on empirical data and industry trends. Success depends on the form of execution. Poor persuaded hybrid models can lead to unnecessary costs, inefficiencies and sustainability trade-offs instead of improvements (Davies, 2018). The potential greenwashing of supply chain initiatives, where companies market sustainability efforts without actual changes, remains as a problem. Retailers must therefore ensure transparent reporting and verifiable sustainable metrics to maintain consumer trust (Nakandala, 2019).

Although hybrid supply chain models have gained increased attention, the existing literature reveals notable gaps. As Jafarian et al. (2020) notes, a lot of the existing research in these areas is based on conceptual frameworks or simulation-based studies, with limited empirical investigation of hybrid supply chains in real-world practices. Particularly in sector-specific applications for instance as grocery retail remains underexplored, especially in relation to how firms operationalize hybrid models in order to manage uncertainty, regional variability and sustainability demands (Jafarian, 2020). Jafarian also argues that there is a lack of studies that examine how decision-makers balance the trade-offs between centralized efficiency and decentralized responsiveness in practice. These limitations suggest a need for more applied,

context-specific research to comprehend the functioning and implications of hybrid supply chains (Jafarian, 2020).

2.4 Triple Bottom Line

2.4.1 Definition and Relevance of the Triple Bottom Line

The Triple Bottom Line (TBL) is a holistic framework to assess the success of a business outside of the traditional way, which was created in 1990 by John Elkington. The traditional way of assessing the success of a business is through only financial measurements. Elkington found a way to measure success by adding environmental and social responsibility to the economic performance. This multi-perspective of the framework then allows companies to measure their environmental, social and economic impacts on a deeper level and will in turn create value among the stakeholders (Correia, 2019).

For achieving sustainability goals that also support business competitiveness the triple bottom line (TBL) framework which includes both economic, environmental and social aspects should be incorporated into supply chain operations. The Swedish grocery retail sector is moving towards a more hybrid supply chain where they prioritize environmentally consciousness such as low emission transportation and efficient logistics (Miemczyk & Luzzini, 2019).

Correia (2019) states that these sustainability efforts which are established within the TBL, not only shows the ethics and morals of a company but also helps to achieve cost savings, increase their brand reputation and build stronger relationships.

Applying TBL into the business will ensure that they align with sustainability objectives which includes the United Nations Sustainable Development Goals (SDGs), which holds strong persuasion in the Scandinavian business practices (Correia, 2019).

Triple bottom line consists of the 3Ps: *planet, people and profit*. This framework is adapted to direct companies to evaluate their business performance through financial measures and their effects on human population alongside the ecological and social concerns (Correia, 2019).

2.4.2 First Pillar: Profit

The first pillar of the TBL is the economic aspect “profit”. The goal of this aspect is to achieve profit and still sustain financial stability both in the short and long term. Cost efficiency acts as a main driver throughout the whole supply chain management process. The cost optimization and

efficiency occurs through the hybrid supply chain for green procurement, energy-efficiency transport, pricing and discounting, and waste reduction. Businesses that put resources into a more sustainable supply chain leads to better brand image, reputation and long-term success (Correia, 2019). Profit does not dismiss the economic performance but it aligns it to a broader economic impact and creation of value. According to Correia (2019), in the start of the 2008 economic crisis, more consumers became increasingly aware of the economic sustainability since the financial instability led to threatening a decrease of their personal stability. This example of the consumer's awareness together with the economic pillar, requires businesses now to create an economic value, not only for shareholders but also for the consumers and community. Businesses should include the economic aspects together with the business impact on the society to fulfill the first pillar in the TBL of economic sustainability, “profit” (Correia, 2019).

2.4.3 Second Pillar: People

The second pillar is the social aspects, “people”. The social aspects in TBL refers to the ethical practices which could include fair trade practices with community engagement to promote social sustainability. Social sustainability focuses on ethical labor, fairtrade, employee well-being and training, and community engagement. Stakeholders in the grocery supply chain build lasting trust and can at the same time promote both ethical work and programs to enhance sustainability in the social aspects (Correia, 2019). Businesses can monitor their suppliers and people working throughout the supply chain at the same time as strengthening the brand image and customer trust (Mimeczyk & Luzzini, 2019).

The increasing awareness of Corporate Social Responsibility (CSR), has gained more attention in corporate operations, where CSR has become central to business strategies, reflecting on an increasing awareness of businesses responsibility to the community and society (Correia, 2019). Research according to Correia (2019) shows the relationship between certain CSR practices with consumers perceptions, and that this relationship can create a more robust public relation strategy.

2.4.4 Third Pillar: Planet

The third pillar of the TBL is the environment “planet”. Businesses need the environmental aspects in TBL to measure the decreases in the negative effects of their activities (Correia, 2019). Focuses mainly of reducing the environmental impact especially in terms of energy usage, water usage, waste management and overall ecological footprint. The growing demand of green products and eco-labelling, as well as minimizing plastic are clear indicators that environmental sustainability is crucial in business strategies to meet the demand as well as regulations. Correia (2019) argues that environmental sustainability is the most crucial part of the TBL, where the economic and social dimensions are secondary. This according to the author is because all human activities today are dependent on the earth's resources, systems and limits (Correia, 2019). Grocery supply chains can benefit from TBL through green logistics and sustainable sourcing according to Hammer & Pivo (2016). The TBL can monitor carbon footprint and optimize resources and ecological effects. Also, the stakeholder participation in different sustainability programs could cut down the ecological effects all along the supply chain (Hammer & Pivo, 2016).

2.4.5 Integrating the Triple Bottom Line

Effectively integrating the Triple Bottom Line (people, planet, profit), requires businesses between the three approaches. A business should find an equilibrium between these three components to in turn succeed through a created value and also build trust with stakeholders, while still bringing societal benefits. This balance between the pillars move businesses beyond the short-term goals and now incorporate sustainability into the strategy core, to create a long-term strategy and value (Correia, 2019).

The research explores hybrid supply chain approaches which tackle expense factors and ecological challenges in the Swedish grocery market. Businesses that use the TBL framework surpass the financial metrics by adding the different elements of the sustainability performances into their supply chain (Hammer & Pivo, 2016). Creating a supply chain strategy that is based on the TBL framework could in turn lead to improved sustainability results, social responsibility, reduction of the environmental impact and constraint risks. This will not only meet the market demands of the increasing awareness of sustainability, but also build long-term profitability and market differentiation (Miemczyk & Luzzini, 2019).

Correia (2019) demonstrates that in order to fully apply the TBL, business will have to go beyond the compliance-based sustainability, which includes sustainable thinking in all of the business practices.

Cross-functional collaboration internally is another important factor in the integration of the TBL. Sustainability is not a responsibility of one single department, but it should be led in all departments such as procurement, operations, marketing, among others. A united view inside the company ensures that sustainability initiatives are not used separately but in an integrated way to adhere to the value chain (Correia, 2019).

Leadership commitment as well as stakeholder involvement is critical for integrating TBL into its business practices. Without a strong leadership that will drive the sustainability topic as a strategic way of evolving the business, efforts are likely to remain superficial (Correia, 2019).

The Triple Bottom Line goes beyond the regular sustainability initiatives where this requires a systemic transformation, a change of mind and embedding in the business core. When implemented properly, TBL not only positively impacts its success but also contributes to the social and environmental well-being, which are fulfilling the wider perspective of a sustainable business (Correia, 2019).

2.5 Cost Drivers and Operational Trade-Offs

2.5.1 Cost Drivers in Supply Chains

In recognizing and managing cost drivers within supply chains, it is essential to understand factors within specific sectors and how they impact the more significant operation efficiency.

According to Holmberg & Örne (2013), transportation is the primary cost in the grocery retail sector. Empty running represents approximately 17% of total kilometers in Swedish road transport, showing notable inefficiencies (Holmberg & Örne, 2013). Cost drivers are the handling costs, another essential part where benefits from full shipments must outweigh the additional costs generated at the distribution hubs (Holmberg & Örne, 2013). Facility operation costs, like maintenance and management of distribution centers, are also cost drivers. Moreover, poorly designed infrastructure, integrated with the challenge of scattered distribution locations, drives up supply chain costs (Holmberg & Örne, 2013). At Coop, return flows include up to 50% of truck capacity, indicating there are also costs related to return logistics (Holmberg & Örne, 2013). Holmberg and Örne (2013) quantify the advantages of horizontal cooperation, writing a

general 6.2% decrease in distribution costs, with variations depending on each corporation's individual conditions. Although, the study meets limitations based on their limited focus only on shared distribution hubs. Suggesting further research in other areas to fully understand and mitigate other potential cost drivers like hidden costs, difficulties with profit sharing, and regional cost differences (Holmberg & Örne, 2013).

Beckeman (2011) addresses a notable change in responsibility between manufacturers and retailers through changed cost structures from their study, allowing retailers to leverage control over their shelf space and their distribution channels. This enables retailers to position their pricing pressure on manufacturers, tweaking margins and focusing on cost reductions instead of value creation (Beckeman, 2011). When there is less cooperation between retailers and manufacturers it has an effect of adding additional transaction costs. The growing popularity of retailers own branded products especially increases the competition by forcing prices down. Thus, brand manufacturers are pressured to find better cost effective ways to make their goods, seeking to maintain high quality while reducing production costs (Beckeman, 2011).

Increased demands for unique products and smaller order sizes put considerable tension on logistics and order fulfillment costs, influencing the efficiency of the operation (Beckeman, 2011). These relationship dynamics create pressure between cost focused transactional processes and value creation over time. When companies in a supply chain struggle to function effectively together and maintain trust, it results in higher costs and limited opportunities for innovation (Beckeman, 2011). Structural inefficiencies and lack of alignment among supply chain members continue to form barriers to achieving optimal costs and enabling innovation in the grocery retail market in Sweden (Beckeman, 2011).

The research from Naga Venkata & Ravilochanan (2014) evaluates that approximately 75% of the cost of a product can be attributed directly to supply chain operations, declaring the significant possibility for cost savings through optimized management (Naga Venkata & Ravilochanan, 2014). These costs represent around 12% to 50% of the total costs through diverse product categories, driving retailers to seek methods aimed at their reduction actively. Required cost reduction methods specified include the elimination of wholesalers through limited distribution channels, direct manufacturer transactions, outsourcing logistics services, and adopting Just-in-Time (JIT) stock methods (Naga Venkata & Ravilochanan, 2014). These activities allow for lower stock-keeping costs and improve warehousing and transportation

efficiency. Cost drivers within SCM (Supply Chain Management), as addressed by Naga Venkata & Ravilochanan (2014), can be mitigated through methods like reducing waste and implementing strategic use of technology to enhance information sharing and forecasting precision. This study primarily focuses on the Indian retail sector, which may restrict the direct application of our use case within the Swedish grocery sector.

2.5.2 Trade-offs in Hybrid Supply Chains: Centralized vs Decentralized

Lots of research exists regarding the trade-offs between centralized efficiency and decentralized responsiveness in hybrid supply chains. One particular study by Validi et al. (2014) has critically reviewed models within the sustainable food supply chain within the dairy sector. Centralized models mainly concentrate on cost minimization through optimized transportation planning and efficient distribution logistics infrastructure. In contrast, decentralized methods accentuate environmental sustainability, especially decreasing CO₂ emissions through environmentally sensitive transportation decisions (Validi et al, 2014).

Further examining more articles, Ghasemy Yaghin and Khalajmehri (2024) address the textile industry sectors centralized and decentralized methods. Their study revealed there is an important tension, where centralizing supply chains usually reaches economies of scale and reduces costs, but at the same time, more often results in increased impacts on the environment because of longer transportation routes. As for decentralizing operations, they reduce the environmental impacts related to transportation while causing higher capital investments and operational expenses (Ghasemy Yaghin & Khalajmehri, 2024).

It is important to balance cost efficiency, environmental sustainability, and social responsibility while continuing research to find challenges in implementing hybrid models and leveraging digital technologies (Ghasemy Yaghin & Khalajmehri, 2024). This study underlines applicable trade-offs we can explore further on within our niche context of Swedish grocery retailers.

2.6 Environmental Sustainability Under Pressure

2.6.1 Sweden's Sustainability Goals & Consumer Pressure

Sweden has made a significant national commitment towards sustainability, aiming to achieve net-zero greenhouse gas emissions by year 2045, in line with the EU Green Deal (Swedish Institute, 2024). Fauré (2019) discovered that Sweden is a net importer of environmental tension, meaning that a significant amount of environmental impact in Swedish consumption occurs in other countries. Food products, beside construction and transport, rank among the highest contributors to greenhouse gas (GHG) emissions, land use and water consumption (Fauré, 2019). In particular, the majority of emissions linked to Swedish consumption happens abroad, especially in China, Russia and other parts of Europe. This suggests that Sweden's sustainability policies should not only focus on local emissions reductions but also reconsider supply chain sustainability and responsible sourcing from involving countries (Fauré, 2019).

Alongside the increasing need for a change according to the Swedish national goals to reach net-zero, environmental sustainability consciousness among consumers has risen, since there now is a need to decrease the ecological footprint. Environmental awareness among consumers is today crucial for grocery retailers in Sweden. The Swedish consumers, just like other consumers in high-income countries, are more likely to consider more sustainable options when it comes to grocery shopping (Deconinck et al, 2025). The majority of the consumers in Sweden or any other high-income countries are more aware of their ecological footprint and try to find more sustainable alternatives in their daily life. This change of pressure on the retailers to provide the customers with more sustainable options supports even more Sweden national sustainability goals (Deconinck et al, 2025).

Labels like organic, locally produced and eco-friendly are some of the most trusted ones, where in that they throw their weight on what people then decide to purchase. This increased awareness among consumers has caused the retailers to adapt their assortment of products. Major chains now feature these sustainable labelled products as well as plant-based alternatives to highlight the environmental responsibility in its product branding (Deconinck et al, 2025).

According to Deconinck et al (2025), consumer confusion often appears when discussing this topic. Customers misunderstand the labels or overestimate the environmental benefits of a certain sustainable product. For instance the products labelled with organic or natural are assumed to be better for the planet, thus have not always the lowest carbon footprint.

Andersson (2015) highlights that many Swedish consumers misinterpret sustainability, connecting it with packaging and organic labeling rather than full-life cycle environmental impact, for instance as production emissions or supply chain sustainability. This sort of knowledge gap means consumers make suboptimal choices, sometimes prioritizing organic foods over lower-emission plant based diets (Andersson, 2015).

The Swedish retailers are required to not only provide sustainable alternatives and assortment but educate the consumers about the real impact (Deconinck et al, 2025). Consumer awareness and pressure is very evident in Swedish grocery stores, from increasing demand for meat substitutes to pressure on the grocery store to cut down on plastic and food waste. Retailers who are able to act upon this consumer awareness are more likely to establish long-term consumer trust and loyalty (Deconinck et al, 2025).

2.6.2 Retailers' Role impacting Consumer Behavior

Agricultural products, in addition to meat and dairy, account for a significant amount of Sweden's environmental impact. According to the findings by Tjärnemo (2015), he mentions that meat production provides substantial GHG emissions, nitrogen cycle disruptions and biodiversity loss. Although, Swedish food retailers are unsure about strongly discouraging meat consumption. They tend to promote organic or locally sourced alternatives, which do not necessarily serve to significantly reduce emissions (Tjärnemo, 2015). Swedish food retailers play a critical role in guiding consumers to more sustainable choices. Gharakhani (2020) says that there is a disconnect between corporate sustainability goals and in-store implementation (Gharakhani, 2020). Retailers set ambitious sustainability goals, but store-level execution differs substantially, often because of financial challenges, employee training gaps and fluctuating consumer demand towards sustainable products.

Food waste is still a major sustainability challenge in Sweden. Jansson (2019) argues that Swedish municipal canteens develop approximately 73000 tonnes of food waste annually, with much of it still consumable. Swedish consumers tend to avoid food that looks imperfect or is close to its expiry date, which leads to more food waste (Jansson, 2019). Eriksson (2016) highlights that supply chain inefficiencies and consumer behaviours greatly contribute to food loss, particularly for short shelf life goods. Retailers have tried to address this with the help of

discounting soon-to-expire products and collaborating with food rescue organizations, but the effectiveness of these measures remains limited by consumer hesitation (Eriksson, 2016).

2.6.3 Systemic Barriers to Environmental Progress

Sweden's Generational Goal, proposed in 2010, targets to eliminate major domestic environmental problems without increasing negative impacts abroad (Fauré, 2019).

However, policy loopholes and weak financial incentives interfere with this goal. Notably, carbon taxes for the food processing and retail sectors remain very low at 0.1% of total operating costs, failing to lead to a real change (Eriksson, 2016). While Swedish consumers express great support for sustainability, price exists as a key factor. Gharakhani (2020) notes that many consumers hesitate to pay a premium for sustainable options, even when they admit to prioritize sustainability in surveys. The contradiction shows how hard it is for retailers to offer sustainable products that are also affordable, while also staying competitive in the market (Gharakhani, 2020). In order to move forward, stakeholders must align economic incentives, policy measures and educational exercises to drive substantial environmental improvements in the Swedish grocery sector (Persson, 2024).

Climate change has become a main concern pressuring environmental behavior and public policy in Sweden. As a global leader in addressing climate challenges, they aim to become the world's first fossil free welfare nation (Månsson, 2016). Sweden's dedication to the Paris Agreement and its ambitious national environmental objectives such as reduced climate impact and sustainable resource use, illustrates the structural efforts to address climate change directly (Månsson, 2016). The government has adopted one generational goal, 16 environmental quality objectives and 24 milestone targets aimed at guiding environmental progression. Moreover, 400 million SEK was distributed to the research and development of environmental technology, to further reflect Sweden's proactive stance on climate change (Månsson, 2016).

Despite the national goals and Sweden's top ranking in the 2016 Country Sustainability Index, Swedish consumers often meet a mental gap between environmental concern and actual behavior. Scientific projections estimate a global temperature increase of 1.8 to 4.0 Celsius in the future coming decades, yet consumers actions remain inconsistent with these risks (Andersson, 2015). Andersson notes that individuals still engage in unsustainable consumption even when aware of climate threats. They tend to defend their habits or expect others, such as the

government or companies to fix the problem. It showcases the physiological complexity of climate change perceptions and proposes that deeper individual engagement is necessary to understand environmental values into everyday sustainable actions (Andersson, 2015).

2.7 Make-To-Stock (MTS) vs. Make-To-Order (MTO)

The grocery retail industry across Sweden and applying the hybrid supply chain methods is to attain sustainable efficiency between the environmental sustainable goals and cost reduction initiatives. Under Make-to-Stock (MTS) production strategy businesses produce products ahead of time according to predicted customer demand. The stored products exist in an inventory waiting for customer demand since they have been produced ahead of time. The MTS method is for predictable items that include staple foods alongside dairy and canned foods. MTS ensures quick access since they become more available so we can avoid delays. Reasonable demand forecasting is playing a vital role to avoid overproduction, through this method uncontrolled output leads to piled up inventory which in turn can increase inventory costs and may result in product waste (Romsdal, 2014).

Make-to-Order (MTO) approaches get produced when customers submit their purchase orders instead of ahead. Specialized goods or customized products typically use this approach as a typical method. MTO can in turn reduce waste since the inventory cost is minimized, since production always is made when increased customer needs. The backside of it is the high costs. MTO production operates with smaller batches alongside customer needs, which lead to production expenses and costs increase since the economies of scale can benefit a decrease as a result. If MTO is implemented into its business strategy, it requires high logistics capability which has control of the raw materials and the customers placing the orders. A supply chain system that combines MTS and MTO allows businesses to find an equilibrium between speed and overall cost efficiency (Romsdal, 2014).

Grocery stores have lately shifted from a traditional make-to-stock (MTS) to a make-to-order (MTO) or combined MTS-MTO approach. This is because MTS focuses on particularly non-perishable items like normal goods, that all have stable demand. The approach would then not work for all items since specialized products or customized needs a make-to-order approach since those orders should be put after the customer needs (Romsdal, 2014).

Combining these two approaches, also gives the retailers the ability to both control the inventory while also controlling the waste to keep it reducing, and still have a good demand responsiveness, but this all requires a combination and great logistics planning to maintain (Romsdal, 2014).

3. Methodology

The third chapter provided an understanding of the thesis chosen methodology and method used. When conducting a thesis, these steps were crucial to ensure a credible analysis. Research trustworthiness and limitations were also discussed in the following chapter alongside the methodology.

3.1 Abductive Research Approach

An abductive research approach is being used for this study to explore how hybrid supply chain models affect cost efficiency, environmental sustainability and social responsibility in Swedish grocery retail. Abductive approach represents repeated processes where theoretical concepts are developed and polished through dynamic interplay with empirical data, enabling new insights to emerge from practical observations and theoretical reflection (Collis & Hussey, 2021). It is especially appropriate for exploratory, qualitative research, where current theories may not fully explain complex, real-world phenomena and where empirical data may reveal unexpected patterns requiring theoretical adjustments (Castleberry, 2018).

In this research, semi-structured interviews with professionals in the Swedish grocery retail sector serve as the main empirical foundation. The following interviews are briefed by theoretical frameworks such as the Triple Bottom Line (TBL), Make-to-Order (MTO) and Make-to-Stock (MTS), as well as concepts of trade-offs, centralization vs decentralization and cost drivers. They remain flexible to capture participants' individual own experiences and perceptions. Insights from these interviews will thereafter be examined and polished, following the flexible and detailed approach illustrated by Castleberry. The thematic analysis applied for the interview data will follow an inductive logic, granting patterns, categories and themes to emerge directly from the participants' responses. It will provide descriptive depth and multiple

layers of meaning in how participants picture and manage sustainability, efficiency and supply chain structure. (Castleberry, 2018).

As data is interpreted, instances where participants' responses either align with or contradicts established frameworks creates opportunities for abductive reasoning. For example, participants might describe how hybrid supply chain models combine elements of both MTO and MTS strategies, particularly when addressing hinders related to short shelf life or fluctuating consumer demand (Soman, 2002). Similarly, participants may explain varying interpretations of sustainability, showcasing how the TBL is applied negatively across organizational contexts. These insights have the capability to reshape how the tension between efficiency and sustainability is comprehended within hybrid supply chains in the retail sector. This type of repeated process between data and theory connects with Sandelowski's aspect that qualitative analysis involves constructing meaning through active engagement with real-world complication (Sandelowski, 1995).

The abductive approach recognizes that terms such as "cost efficiency" and "sustainability" are clarified differently depending on organizational priorities, industry challenges and contextual factors (Sandelowski, 1995). The broader context of technology and sustainability evolution in grocery retail supports the usage of abductive reasoning. Frameworks such as the TBL, MTO and MTS are valuable and relevant, but they do not secure the full range of experiences, technologies or strategic decisions shaping hybrid supply chains. As noted by Rantanen in their case study of Finnish grocery retail, abductive research allows for a grounded but also flexible exploration of supply chain cooperation and integration (Rantanen, 2017). Hove-Sibanda also tells how qualitative, abductive approaches can declare how companies acknowledge operational risks, innovation pressures and shifting performance expectations - insights that are very rarely predicted by theory alone (Hove-Sibanda, 2020).

Ultimately, this research focuses to reveal how Swedish grocery retailers grasp and navigate the complications of balancing sustainability goals with cost-driven supply chain decisions. The theoretical frameworks will prepare structure and guidance, but it is the abductive process that will enable this study to evolve the frameworks based on participants' mentioned experiences and practical challenges.

3.2 Research Philosophy

The following study carries a qualitative research design, based on an interpretivist perspective, to discover how hybrid supply chain models determine cost drivers, environmental sustainability and social dimensions in grocery retail in Sweden. Interpretivism enables a broad understanding of how supply chain professionals grasp and navigate the trade-offs among cost drivers and sustainability. The study recognizes that the trade-offs are influenced by company strategies, regulatory frameworks, context-dependent and technological developments (Nickerson, 2024). Given the complexity of hybrid supply chains, thematic analysis serves as the main method for valid qualitative data. The research is shaped through the interaction between researcher and participants, where insights are gathered through understanding, not just by uncovering objective facts. As Sandelowski (1995) emphasizes, qualitative research requires the analysts to actively engage with the data, clarifying findings in a way that remains faithful to participants' perspectives during contribution to theoretical development. Therefore, providing insights that are grounded in real-world experiences and decision making processes (Sandelowski, 1995).

3.3 Data Collection

The process of gathering primary data was done through the execution of semi structured interviews as addressed by Collis and Hussey (2021), with experts, all holding unique positions at competitive corporations in the sector. This provided the research with important practical understandings of real world applications of hybrid supply chain models.

The participants interviewed have higher executive leader roles, grocery store floor-workers responsible for specific sections, as well as store managers. They were chosen to strengthen a mixed perspective from the larger supply chain operation. Participants were recruited through a blend of outreach to organizations, and existing professional relationships. The use of a systematic approach ensured that all participants met the set selection criteria regardless of the two methods.

Interviews were conducted remotely using the digital platform Microsoft Teams, simplifying the process of interviewing and gathering perspectives across different regions in Sweden. An interview guide was created to make sure a clear structure was put in place, making sure relevancy to the literature review. By having a semi structured format, flexibility to explore more unique insights based on each participant's background and experience was a natural process.

They all lasted for around 50 minutes each and were recorded and automatically transcribed with clear approval from the interviewees (Collis & Hussey, 2021). All interviews were conducted in accordance with strict set ethical guidelines, including data storing on university networks, informed consent and confidentiality measures.

3.4 Data Analysis

The interview data will be analyzed through thematic analysis, as explained by Castleberry (2018), beginning with anonymizing the current state of the transcript before importing it into Microsoft Excel. The transcripts will firstly be translated from Swedish into English. We will go through the transcripts multiple times, correcting minor spelling errors within the transcript to gain a complete understanding of the content. We will then move to the breaking down phase by systematically coding the data to identify patterns related to hybrid supply chains, cost drivers, and sustainability practices, creating what is described as meaningful groupings (Castleberry, 2018).

This analysis will progress to the reconstructing step, where transcripts will be converted into codes, where those being related will be connected to create primary themes focusing on, in our case, centralized and decentralized tradeoffs. These will be reviewed against the original data to ensure accuracy. In the understanding state also described by Castleberry, we will define themes and examine their relationships to our theoretical frameworks, specifically the Triple Bottom Line and MTO/MTS strategies. Throughout this analytical process, we will maintain our abductive approach, allowing for ongoing improvement between empirical data and theoretical constructs rather than forcing data into predetermined categories, as data are not in or of themselves explanatory, it requires interpretive work to develop contextually grounded insights (Castleberry, 2018).

The transcripts from each interview were inspected individually, and the most meaningful information was extracted into individual quotes, filtered, and placed within a Google Sheets file as a program of choice. Each quote was broken down and simplified to its intended meaning without altering the original content. These codes started the foundation for open coding (Collis & Hussey, 2021). The codes were grouped through axial comparison (Collis & Hussey, 2021), twenty-two categories in total. These categories were each linked to one main theme based on each code's context. The themes created became Economic, Environmental, and Social, linking

to the three pillars within the Triple Bottom Line framework. To ensure a representative observation, the three most frequently used categories within each theme will be used to develop an extended model of the TBL Framework. These nine categories define the situations raised by each participant and represent how the current operations work.

3.5 Trustworthiness

Qualitative research studies require trustworthiness assessment to establish a research quality and a certain credibility level. Trustworthiness evaluation in this qualitative research occurs through the assessment of dependability, credibility, confirmability and transferability according to Collins and Hussey (2021). The research finding quality depends heavily on these separate concepts to give both reliability and integrity of the interviews.

3.5.1 Dependability

Dependability has a meaning of whether the thesis process is systemic and well-documented (Collis & Hussey, 2021) and is also similar to reliability. It refers to the stability and capacity to be followed and documented correctly. In this thesis case it is in terms of maintaining detailed documentation of the thesis data collection process and how sources and interviewees were chosen. The thesis demonstrates dependability by documenting an extended list of research from the interview questioning techniques to the participant selection strategy as well as the data collection practices to the analysis. The research methodology matches the thesis goal and is explained in detail, to reinforce the dependability of the findings.

3.5.2 Credibility

The meaning behind credibility is if the thesis was correctly identified and described. It shows the accuracy and truthfulness of the data and interpretations. The research shows authentic views of the interviews because of this method. When this thesis was studied, several measures were put into practice like data triangulation in combination with member checking, as well as using different and multiple data sources. Member checking includes sending the interview transcript and findings back to the participant for them to verify the accuracy of the findings along with an opportunity for them to change or give feedback to correct misinterpretations. Data triangulation is an important aspect in credibility. Data triangulation strengthens the credibility or

trustworthiness by using multiple sources, and confirming that the information gathered is not biased (Collis & Hussey, 2021). When the interviews were conducted the thesis underlined the interviews from the employees and owners, company policies regarding different aspects both social, environmental and economic.

3.5.3 Transferability

If the findings of the thesis can be applied to other situations which also connects with external validity and generalizability. Through qualitative research this concept avoids broad generalization seeking it to remain vital to provide enough material which enables the reader to evaluate how the thesis findings might be applied to similar settings. Different charts and tables are used in this thesis to enrich the data used and to support the readers understanding of transferability based on their judgement. This thesis uses specific, chosen secondary data sources and interviews and to ensure that transferability is done the thesis will use detailed explanation of context, pictures and settings (Collis & Hussey, 2021).

3.5.4 Confirmability

Confirmability is the same as objectivity and incorporates findings based on data and not bias. All steps of data analysis require a transparent documentation system which includes coding procedures, documentations and analysis throughout the whole thesis analysis process. Direct quotations from participants in both the coding and the thesis text, act as a key feature in this thesis because it raises the participants' voices directly to the reader. The findings receive validation through cross-checking because the collected data undergoes comparison to the literature to ensure alignment between the collected data points (Collis & Hussey, 2021).

3.6 Ethical Considerations

Ethical considerations refers to the meaning of forming a code of conduct based on the moral and values the thesis has (Collis & Hussey, 2021).

To maintain the highest and most secure level of ethical considerations, all the participants and interviewees were voluntarily participating in this study. This means that the thesis did not involve any forced interviews, and all was voluntarily participating (Collis & Hussey, 2021). The interviewees were provided with information and details about the thesis purpose and vision, and

also about the interview process. After being given the information they were asked to either accept or deny the participation by signing a study consent form which is aligned with voluntary participation and ethical considerations (Collis & Hussey, 2021).

These interviews were made through teams-meetings since all the participants live in various cities in Sweden. In teams there is a setting to transcript the meeting, which we did. The participants were also informed that the meeting was going to be transcribed. To meet the GDPR standard and guidelines during the interviews and especially the transcription which is personal data being stored, we considered the privacy and protection of all the participants' identities.

The thesis is going to refer to each interviewee by different roles or names like “participant 1” to align with either anonymity and confidentiality. Anonymity refers to the meaning of the assurance that all participants will be anonymous with their names, exact roles in what company, age and gender. Confidentiality instead refers to that participants that provide the information can not be traceable to the organization providing it (Collis & Hussey, 2021).

The participants will not be referred to by their names, exact roles in which company, age, gender, or other personal information that could identify them. The participants will be named as Participant A-G, and by using these coding names, it reduces the risk of the ability to identify, the interviewees may be more open to talk and discuss different topics without any names or identities used. The participants could also draw back their interview if wanted.

All of these guidelines were discussed and informed about which is crucial in terms of expressing transparency and building trust towards the participants to make them feel safe in discussing their work with us.

3.7 Limitations

When working with credibility the thesis underlined the method of triangulation to enhance trustworthiness and not be biased. The thesis incorporates both interviews with employees but also company policy documents.

A key limitation is between the goals outlined in example ICA Gruppen Policy documents and the observed practices were seen. ICA Policy aims to have all of their food packaging material for their own label emerge as reusable or recyclable by 2022 (ICA Gruppen, 2024). During the interviews we observed that non-recyclable materials such as plastic gloves were used widely at work and has no plan right now to be removed due to the freshness and cleanliness as plastic

gloves give. Although many of their packaging follows this, it also shows that the overall goal and internal-use products do not benefit from the packaging policies even though their sustainability commitments is to reduce overall plastic usage. It is hard to align all the policies in their daily practices, and especially those who benefit. Nonetheless, it is important to compare policy documents with the interview answers to reduce the risk of bias or favoritism towards their own store, and instead give a more balanced view when comparing.

One thing that could have been done to strengthen the triangulation of credibility more is to observe team meetings and not only focus on the interviews and policies (Collis & Hussey, 2021). The limitations of not directly observing the real work or in team meetings lead to lack of behavioral data, where real-time scenarios are limited. Potential response bias could impact the interview where the employee could express their experiences in a more favorable or worse way than it is, and when using cross-checking the trustworthiness would have helped. The lack of using multiple perspectives as an observation, there was an opportunity missed on analyzing their interviews answers compared to their actual behavior in the workplace. Future research could incorporate one more method to increase the credibility and validate the understanding of the dynamics in the workplace.

4. Findings

The fourth chapter included the empirical findings derived from the seven interviews conducted. The purpose of chapter four was to summarize and analyse the findings, and strengthen this analysis by the interviews direct quotations, to create a more stable and trustworthy analysis.

The following chapter derives an analysis of information from the semi-structured interviews conducted with the seven participants. All participants received anonymous status in the thesis analysis and refer to them by letters from A-G, to preserve their confidentiality and safety throughout the whole process. 26 questions were asked to the floor-workers, and 29 standard questions to the executive leaders. The interview was conducted and transcribed in Swedish, since the thesis explored the Swedish grocery stores. There were also in some cases added additional questions when needed during certain parts of the interview. The responses from the interviews are both summarized and analyzed below.

4.1 Economic

The findings from the thematic coding and interview uncovered a wide range of perspectives but especially the economic. Reflecting on the interview answers it becomes clear that the economic perspective with price constraints is dominant in the business strategy.

Participant C started by stating that their organization uses a centralized approach for handling frozen goods. The centralized model focuses next on regional distribution as C explained *“the first thing is that we have our terminals, which are centralized. This is for the central assortment that is stocked. It is transported to so-called transshipment terminals. We have twelve in Sweden”*.

This system with centralized model works best for short-term products since participant C also explained that they have a rule of thumb for items to remain half of their shelf life until reaching the stores. This shelf-life needs retailers to choose cross-docking or make-to-order operations because it helps maintain product freshness throughout the whole delivery process until it reaches the store.

Through cross-docking facilities perform efficient delivery routes for their items. Participant C explained that *“cross-docking suits niche or local low-volume goods”* which explains that such items lack enough value to justify their placement on the shelves, and then instead you deliver niche products outside the major cities since the request is not that high.

A wide range of delivery schedules exist across different locations throughout the retailer's stores and departments. The delicacy section receives numerous daily deliveries because they purchase their products from multiple supply companies according to participant B. On the other hand, participant D explained *“we get deliveries six days a week, which means we run pretty high”*.

On the other hand, delivery disruption is standard practice. Participant D continues with *“we always get a notice from the central hub if delivery is late. If it's one or two hours late we have a plan on how to redirect the day”*.

The deliveries are unpredictable and according to participant G *“the dairy yesterday brought five pallets and today only one came, so it's very shaky”*.

All of the participants expressed and shared similar opinions of the demand forecasting. They share the use of tools to calculate upcoming orders, but also their opinions on how to increase

orders based on historical and analyzing upcoming forecasting. Participant D noted that *“we try to go a lot on forecast and that's what we have to look at”*.

More advanced methods of demand forecasting were seen but not as common. Automated systems are increasing in these stores and as participant D explained *“we have a system called Automatic Order Placement where you put in a forecast and then get suggestions on how much to order”*. But still, most of the stores are calculating their upcoming demand forecast by hand, and are using their own knowledge to order, which participant B explained as *“order tools are hard, we go by binder, check stock by hand, order by phone”*. The development is growing, and it will ease the ordering process, but many stores are still doing it by hand.

Problems could arise when doing demand forecasting and order by hand since it is often hard to predict. Factors that could arise from predicting demand is changing weather that impacts the purchasing decisions as participant C stated, also crises like COVID since it was hoarding periods, but also crisis like backordering at suppliers as A explained. All of the participants had divided processes of ordering and keeping stock levels, but they shared the same experiences of factors that could cause problems when ordering both by hand and by computer. These challenges are hard to predict and know beforehand, so even though demand is forecasting it is hard to avoid these.

The technological adaptation is an area of growing interest in the stores which can transform the whole retail sector. There were divided responses, but all of the stores are using some kind of technology but can be both handheld phones to order from, or more advanced terminals. One of the most advanced technology developments is at participant C's department where explained *“to achieve maximum efficiency we have a centralized model, which for example has allowed us to build one of Europe's largest fully automated terminals for groceries and food”*. As well as the development of technology and innovation, AI is emerging. These AI solutions are as well as the innovation increasing but not fully adapted.

4.2 Environmental

The present market shows an increasing trend towards plastic reduction which faces organizational limitations. According to Participant B the grocery store transitioned from plastic boxed to cardboard several years ago yet complete changes and impossible at present like their

use of plastic gloves. All the participants had a wide usage of plastic gloves that can not be replaced, based on the freshness and quality it brings when handling food.

Specialized packaging programs have been introduced by various stores across the departments. Based on participant D *“we’ve changed, one could say in the ready-meal shelf, we’ve switched to all bamboo trays now that exist”*. This is common throughout all of the interviews, that they have had shared thoughts on the reduction of plastic and use reusable materials instead, which is also reflected on their policy views, and that is one part of why they have such common opinions too. However, customer preferences still conflict with these new policy strategies of reducing plastic. Although customers care more about sustainability but according to participant A she told us *“many customers still want the fresh fish wrapped in plastic so it doesn’t leak, so we use a lot of plastic still”*, but according to participant F, *“for packaging there’s basically no plastic, most cardboard”* and this is seen through all of the interviews done.

These organizations work together with other industries to implement more reusable packaging systems. Participant C explained that their industry *“operates together as co-owners of ICA and Axfood, Lidl and Svenska Retursystem, where plastic pallets and plastic crates are produced by the company for reuse purposes”*.

The reduction of waste heavily relies on markdown or discount strategies for most of them. Participant A said *“for food waste we also have the discount box where everything expiring today or tomorrow is marked down so we can sell instead of binning”*. This approach is shared for all of the grocery stores where they all mark down goods that customers then can buy at a discount price, which turn into less food waste. On the other hand, if food gets returned or customers make a claim on them they can not process it or sell it and then it goes straight to food waste according to participant D. According to participant E, their store *“donates food to Hela Människan. It is an organization that picks up food that still is edible even after its past expiry date”*. Foods that are on discounts or special prices are often close to their expiry date but have not yet expired, but when they are, they are often still edible. Participants E stores then think it is better to donate this edible food than throwing it away, and this also gives them an advantage to the other participants, since they are the only ones focusing mostly on donating. At the same time participant G explained *“we can’t donate expired products, there’s some law against it”*, so they can not donate food according to their policy and regulations and are then forced to in the end throw it away. The other participants focus heavily on getting the food sold on special prices.

Most of the grocery stores based on the interviews have structured systems, both by hand and by computer where they monitor their waste. Participant F explains that *“we are supposed to register all the waste, and every week we look at the percentage of food waste based on the registrations”*, and this together with participant D who explained that the process of food waste management occurs by weekly and daily monitoring with controls.

Renewable energy investments are significant. Participant C mentioned that *“our terminal that is built is environmentally classified, environmental class silver”*, but also *“we have 38.000 square meters of solar panels on the automation terminal and about 5.000 on the freezer”*. This terminal benefits from its recycling operations and renewable energy since they recover heat during cooling processes.

System trains that are in operation between Malmö and Eskilstuna diminish the environmental emission according to participant C. *“We have two system trains that run between Malmö and Eskilstuna. They take deliveries from suppliers that produce in southern Sweden or are important, and then to the terminal in Eskilstuna, which go back with store-packed goods to the stores in southern Sweden”*, this implicates a step further in the renewable energy sector, where their store really are recycling the energy and uses themselves of renewable energy instead of fossil fuel trucks. Participant C showed the importance of using their system trains instead of fossil fuel trucks by *“it enables us to reduce delivery days to the store and that is, how many times the transport goes back to the store, which benefits both environment and cost”*. This change to system trains is important for the upcoming future, and changing to these initiatives step by step is a way further than the other participants' stores or departments. Participant D focuses instead on *“fully loading the trucks on each delivery”*. So, they share a common goal against working for the environment and reducing emissions but in different ways and ambitions. The majority of the stores use written environmental targets as a big part of their operations. The yearly environmental goals revision process as participant D centers around *“7-8 different environmental goals that we revise every year”*. These goals for the departments include long-term objectives to replace fossil fuel to renewable, from plastic to cardboard and so on as seen throughout the interviews. These environmental goals could also be taught with education. Participant A explains that *“education covers the environment plus handling raw materials, required temperatures, where environment is included”*. This shows that education is a central

part for them in turn understanding the goals but also working towards completing them, since the employees get all the information needed why its necessary to align with sustainability goals.

4.3 Social

The social part was also dominant in the interview where focus laid on supplier selection, local vs international but also about customer feedback and compliances.

A major strategy that all of them have is the local products. Participants A, C, D, E, F and G explained its importance for them to have local food and G also expressed the appreciation from customers when they bring in locally produced food. Participant D said *“Then we also work a lot with local suppliers and want to work a lot with local suppliers because that’s very important”*, which expresses the importance of local suppliers for the social perspective and to benefit the locals. Participant G pointed out that *“It is usually small suppliers. There can be a strong local connection, what we usually call honey suppliers. They do not have the ability to produce the item in sufficient volume; it becomes too expensive to stock it nationally. If it is only sold locally, it becomes much cheaper transport-wise. There is a connection to cross-docking”*. This quote is showing that local production requires a strong local connection where it is often easier and less expensive to transport from.

At the same time the choosing of the local products is based on the increasing customer demand for it as said in the interview with participant A, *“we have a lot of local and Swedish goods, but I think it’s because customers want Swedish, locally produced, not mainly for transport reasons”*. At the same time participant G compared it to when this person worked in Rissne compared to where he is today, that these two are a contrast to each other and especially when choosing organic and local food. Participant G explained *“there are both types, and when I worked in Rissne, there wasn’t so much organic stuff there, but here they are more aware and do not care about pricing”*. This clearly shows that local and organic food works better in different areas, and they might need to adapt their ordering based on the customer behavior. The customer behavior and ordering pattern is closely related and the customer demand for local products may be increasing in some, and decreasing for the others.

The social aspect in terms of compliance and regulations are focused on. There are a lot of regulations, certifications and laws to have control of when introducing a new product into the market, but also to keep up with the products in store as participant A said *“we can't sell if not labelled, it's strict. Last autumn prawns lost the MSC label, and we weren't allowed to sell them until they came back”*. The example participant A gave was important for the understanding to not only keep up with the new products, but to control and have information about the past and present ones. Participant C mentioned the number of measurements of sustainability targets they have which is driven by regulatory requirements and work with CSRD.

Although they have information about the products, most of the suppliers are chosen by the top management at their grocery stores which means that they have approved it somehow as participant G explains.

The certifications and requirements alongside social sustainability are taught to their employees and according to participant C *“To secure competence and motivation, it is important to communicate the target picture and the desired effects. We also have individual conversations with each employee where previous years are followed up and graded with reflections on work environment and well-being.”* and they are *“measuring employees' perceptions in pulse measurements four times per year”*. These quotes are showing the motivation from the top management team at their local grocery store to increase employees competence for them to in turn get more knowledge. Courses are published on their internal site by both participants G and E, which includes courses and tests not only on environmental sustainability but also other sustainability aspects. The published required courses are the certain basic skills to know, and there is also training to do when starting at the job. This focuses on the educational opportunities the stores have to develop the employees personal as well as work skills.

5. Qualitative Analysis

The fifth chapter introduced the qualitative analysis of the empirical findings in relation to the theoretical framework. The purpose of this chapter was to interpret and bridge the interview data with the literature, identifying patterns, trade-offs and strategies that addressed the

research question. The analysis was structured thematically to provide a summarized understanding of how Swedish grocery retailers applied hybrid models in practice.

5.1 Hybrid Supply Chain Configuration

The interview data strongly establish the presence of hybrid supply chain configurations among Swedish grocery retailers. It agrees with what Liu (2024) and Vallandingham (2018) described about hybrid models combining centralized efficiency with decentralized responsiveness to handle growing complexity in food logistics. They highlight the growing importance of such models in grocery retail because of increasing consumer demands, perishability of goods and the urgency of technological integration. As Lin (2019) argued, centralized features such as unified inventory control and planning with decentralized operations such as regional fulfillment and local sourcing, allow for flexibility and resilience. A key insight from the interviews was that retailers use centralized terminals to provide regional transshipment hubs, a structure that copies the centralized component of hybrid supply chains explained in the literature. As declared in Section 4.1, Participant C explains a system where centrally stocked goods are routed within twelve transshipment terminals, permitting for integration of local suppliers with cross-docking. This dual-functionality aligns with Vallandingham's (2018) idea of combining centralized infrastructure with decentralized execution, which he notes is pivotal for handling growing consumer demands and regional preferences in grocery retail. Moreover, the model supports cost control through streamlining inbound logistics and linking local supply without adding warehousing complexity, a benefit also mentioned by Nakandala (2019), who stresses the importance of execution efficiency in hybrid supply chain network design. This sort of setup also mirrors the broader trend described by Liu (2024), where hybrid supply chains are adopted in order to balance the cost advantages of centralization with the flexibility and sustainability of localized operations. Additionally, this structure connects with the Swedish market context, where logistical reach across sparsely populated regions for instance as northern Sweden can be extended cost-effectively through transshipment points, strengthening both service coverage and environmental goals. Such configurations are only viable with the support of enabling technologies like real-time tracking and demand forecasting, which Vallandingham (2018)

identifies as key components for optimizing hybrid models under complex supply and demand conditions.

The hybrid design also reacts to product perishability and shelf-life constraints. Participant C noted that products must arrive in-store with at least half their shelf life remaining, which can be challenging for long-distance deliveries. To solve this, they use cross-docking and make-to-order methods to further extend freshness and responsiveness. This response directly supports Romsdal's (2014) argument that integrating Make-to-Order strategies with Make-to-Stock systems supports in reducing waste and response to variable demand, especially in hybrid systems where product shelf life and lead times are linked narrowly. Participant C also revealed that retailers use cross-docking to bypass warehousing for niche or low-volume items, particularly those requested by stores in rural areas. These products tend to lack the sales volume to advocate for centralized stocking, making direct transshipment a more adequate and cost-effective solution. It shortens lead times from production to shelf, improving responsiveness while minimizing storage and handling costs. This sort of action aligns with what Liu (2024) was discussing, how the hybrid systems provide local responsiveness while avoiding the high operational cost of stocking niche goods centrally.

Transshipment terminals were also described by Participant C as expanding reach without influencing cost, which again aligns with Nakandala's (2019) take on execution and network design *"If we set up a transshipment terminal in Luleå, a supplier can deliver to that terminal in Luleå. Then it goes all the way up to the national border and the Finnish border and the Norwegian border. It is a huge area. But it does not cost more, because an extra truck is not driven to deliver it to those stores."* This logistical structure symbolizes what Nakandala (2019) identifies as critical to effective hybrid supply chain execution: minimizing transportation redundancy while permitting local responsiveness. It also showcases what Liu (2024) terms like "low-carbon optimization," where strategically located hubs decrease both cost and emissions by shortening final delivery routes. Additionally, this practice supports the environmentally conscious logistics highlighted by Vallandingham (2018), who emphasizes the importance of data-driven supply chain planning by using tools like AI-based forecasting and real-time routing in order to improve reach without sacrificing sustainability goals. From a strategic standpoint, it

aligns with Romsdal's (2014) argument that hybrid supply chains must be configured for efficiency but also to respond to demand volatility and perishability, especially for grocery items with limited shelf life. The mash-up of centralized planning, regional transshipment and local adaptability shows that Swedish grocery retailers are not only experimenting with hybrid models, they are operationalizing them in ways that are refined for their unique geographical, environmental, and product-related challenges. It fortifies that hybrid supply chains are not static structures but instead dynamic systems that need frequent refinement situated on empirical demand patterns, infrastructure capacity and sustainability mandates.

Beyond its structural efficiency, the hybrid supply chain model provides several key benefits for grocery retailers. It enables more rapid response to local demand fluctuations through decentralized ordering, while centralized hubs minimize logistic costs and streamline bulk purchasing. The freshness is improved via cross-docking and localized sourcing, which reduce delivery times for perishable items. Also, the model strengthens supply chain resilience by diversifying fulfillment nodes, decreasing the impact of disruptions in one part of the network. These benefits connect with Lin (2019), who says that combining centralized oversight with localized execution improves operational continuity and responsiveness in dynamic environments.

5.2 Cost Drivers and Operational Trade-offs

The found data in the interviews reinforce the literature's view that transportation, inventory management and purchasing processes are vital cost drivers in the grocery retail supply chain. The cost-related pressures have to be balanced against responsiveness, service quality and sustainability. These are central concerns in hybrid supply chain models. In particular, transportation costs were continuously referenced as pivotal to cost efficiency, which agrees with Holmberg & Örne (2013) findings that transport represents a great proportion of grocery logistics costs in Sweden. Participant D addressed how their organization consolidates delivery routes to maximize truck utilization and avoid sending out underloaded vehicles. If there are insufficient heavy goods for a delivery, they wait with the shipment to the next day to do fuller loads. This practice minimizes empty runs and improves transport efficiency, directly addressing a major cost driver in grocery logistics. This type of practice mirrors the need to manage "empty running", pinpointed by Holmberg & Örne (2013) as a utmost inefficiency. Retailers know of

this and adopt volume-based delivery timing strategies to minimize wasted capacity and fuel usage.

The data also notes delivery frequency trade-offs, which showcase broader tensions in hybrid systems between centralized efficiency and local service needs. As previously noted in the findings, Participant D described a delivery schedule of six days a week, which requires tight inventory management despite having a large back-store. This reflects a deliberate strategy to minimize holding costs while still maintaining high service levels, emphasizing the operational tension between lean inventory practices and consistent product availability. It basically illustrates an effort to reduce holding costs while also maintaining service levels, which is a core tension in cost-efficient, lean operations. As Beckerman (2011) mentions, retailers increasingly control logistics planning in order to optimize shelf space and avoid unnecessary capital tied up in inventory. Another operational cost driver is inventory accuracy and ordering strategy, with interviews illustrating both manual and automated approaches. As briefed in the previous chapter, Participant B said that stock levels are tracked manually using binders and orders are placed by phone, without the support of historical sales data. This traditional, experience-based approach contrasts with automated inventory systems and can lead to inefficiencies or inaccuracies in demand forecasting and replenishment. It stands in contrast to automated systems like computer-assisted ordering, to minimize labor costs and increase precision. Although, even stores with automation see challenges, as mentioned by Participant G *“Most of it comes on auto-order. And for that to work I have to have the right stock checks. That is, the right stock levels in the store.”*. It illustrates a critical trade-off introduced by automation. Although it decreases manual workload and can streamline inventory management, it also introduces a dependency on data accuracy. If the system’s input data like the stock levels is incorrect, it can lead to stockouts, overstocking and in time financial or environmental waste. Vallandingham (2018) explained that automated systems in hybrid supply chains are only as effective as the quality of the data and human-system coordination that supports them. He warns that without daily updates from store-level staff and connection between digital systems and real-world practices, automation will most likely lead to failure to deliver its promised benefits. In other words, the efficiency gained through decreased manual input could come at the cost of increased vulnerability to error, evolving the need for investment in employee training, accurate inventory tracking, and integration between operational layers.

Purchasing decisions are also presented as a pivotal driver of direct and indirect costs. Inaccurate purchasing inflates inventory levels but might lead to unsellable goods or waste. As Participant D said *“Purchasing is a big thing that can create many problems if it goes wrong.”*. It aligns with Beckerman’s (2011) findings that poor coordination within supply chain actors, such as hybrid models, results in additional transaction costs and inefficiencies. It also reflects with Naga Venkata & Ravilochanan (2014), who found that up to 75% of a product's cost can be linked to supply chain decisions, like purchasing, logistics and fulfillment strategies.

Trade-offs were in particular visible in the Make-to-Stock vs Make-to-Order strategy. Participant C acknowledged that different types of products need different inventory strategies to manage cost and waste. *“For goods with a short shelf life we take in orders from about thirty suppliers who produce exactly what the store has ordered. It is produced and delivered to the central terminal where it is quickly put in stock and immediately picked, then added to the store order together with the other goods already in the terminal.”*. This reflects with Romsdal’s (2014) recommendations for using MTO in cases of volatile or short-shelf life demand, while MTS remains suitable for predictable, high-turnover goods. The combination of both strategies allows stores to reduce waste while maintaining availability, which is a hallmark of hybrid supply chain flexibility.

5.3 Sustainability Integration via Triple Bottom Line

5.3.1 Economic Dimension

The merging of sustainability into supply chain operations was a continuous topic in the interviews and connects closely with the Triple Bottom Line framework. Multiple participants talked about the importance of managing costs through improved forecasting, inventory control and purchasing accuracy. These sort of strategies mirror the “profit” pillar of TBL. Participant D explained the direct connection between purchasing decisions and waste *“Because purchasing errors often lead to waste, which usually leads to us throwing away food and that becomes food waste we don’t want.”* and *“Purchasing errors can also mean we hold the goods too long in stock and it becomes expensive inventory, so purchasing is the big issue that we work a lot more with and I think we have...”*. Participant C also noted the struggle of transporting small volumes *“If a small volume needs to go to a store, it becomes very expensive to make it happen.”*. These quotes mirror the operational trade-offs discussed by Holmberg & Örne (2013), who notes that

inefficiencies in transportation planning and forecasting raise logistics costs but also contribute to environmental burdens through fuel consumption and unnecessary waste. It supports the broader idea that sustainability efforts in grocery supply chains are not limited to environmental metrics alone, but are closely tied to cost-efficiency and process optimization. Focusing on better demand planning and purchasing accuracy, Swedish retailers aim to simultaneously decrease food waste, improve financial margins and strengthen logistical efficiency in order to demonstrate how the "profit" pillar of TBL can align with environmental and operational goals when embedded into supply chain strategy.

5.3.2 Environmental Dimension

The respondents also talked about the actions taken to decrease environmental impact, aligning with the "planet" pillar of the TBL. Participant D said that they had made a shift to more sustainable packaging materials, like beige bamboo trays. These can be less visually appealing for sales purposes, but are considered highly more environmentally friendly, postining with broader sustainability goals. In terms of circular systems, Participant C explained that their organization, along with other major retailers like ICA, Axfood, and Lidl, co-owns Svenska Retursystem, which is a company responsible for producing reusable plastic pallets and crates. These are circulated across stores nationwide, cleaned, and reused in a closed-loop system, illustrating a coordinated industry effort to reduce waste and support circular logistics practices. These practices align with Liu (2024) and Zhou (2018), who emphasize the role of closed-loop systems, for instance as reusable transport packaging and consolidated logistics flows. These are essential strategies within hybrid supply chains to minimize emissions, lower material consumptions and meet consumer and regulatory expectations for sustainable operations. Liu (2024) explains the operational value of integrating low-carbon practices into supply chain design, while Zhou (2018) focuses on the importance of circular resource usage and environmental responsibility in supplier selection and logistics management. The use of systems such as the Svenska Retursystem illustrates how Swedish retailers are integrating these principles by reusing pallets and crates across national operations in order to minimize waste and environmental footprint.

5.3.3 Social Dimension

For social sustainability, the “people” pillar of TBL, was most evident in how companies enforce supplier responsibility and employee care. Participant C said the following: *“We have a policy for suppliers that they must sign called Code of Conduct, and there a lot about social sustainability is regulated—child labor, for example, or compliance with laws and regulations.”* and also mentioned that their organization conducts pulse surveys four times a year to measure employee perceptions. This practice symbolizes an ongoing commitment towards internal social sustainability by regularly monitoring staff well-being and engagement as briefed in the findings chapter.. It relates with Miemczyk & Luzzini’s (2019) take on ethical labor conditions and supplier accountability as foundational to socially sustainable supply chains. They say that ensuring fair treatment, legal compliance and social responsibility throughout the supplier network is crucial for long-term stakeholder trust and operational resilience. At the same time, Hammer & Pivo (2016) expands the definition of social sustainability to include internal dimensions for instance as employee engagement, workplace well-being and organizational culture. According to them, companies have to monitor supplier practices but also foster healthy, safe and inclusive working environments within their own operations. The pulse measurements explained by Participant C showcases this internal commitment, suggesting that social sustainability is not solely a compliance issue, but as an active, ongoing internal priority.

5.4 Aligning with MTS/MTO Strategy

Swedish grocery retailers frequently apply both Make-to-Stock and Make-To-Order strategies relying on the product type, demand variability, and shelf life, which aligns tightly with Romsdal’s (2014) argument that a hybrid MTS-MTO approach supports both efficiency and flexibility in the grocery sector. MTS is mainly used for high-volume, predictable goods, while MTO is used for seasonal items, short shelf life products and customer specific orders. These models are adapted to decrease overstock, manage perishability and meet changing consumer needs, especially in hybrid supply chain structures where central and decentralized processes coexist. As previously mentioned in Section 5.2, Participant C, who described how the organization manages perishable goods using a just-in-time logic. Instead of holding stock, the retailer coordinates directly with approximately thirty suppliers to produce and deliver only what is needed based on store orders. These items are swiftly processed through the central terminal

and integrated into the broader delivery flow. This MTO approach allows for minimal stockholding of volatile items and helps preserve shelf life. As Romsdal (2014) notes, this strategy supports sustainability by reducing food waste and unnecessary storage, while also avoiding excess transportation created by overstock movement. In contrast, MTS is relied on for standard or more predictable goods. As Participant A says *“In the fish department we have a basic assortment we always keep every week.”*. It shows the role of MTS in ensuring product availability and quick replenishments, particularly for staple goods where demand is stable. Although, it needs accurate forecasting to avoid overstock and spoilage, importantly in fresh product categories. As Participant G explains *“We go a lot by the statistics. But it also depends... you have to mix gut feeling with statistics.”*. Forecasting challenges were further emphasized during seasonal peaks, where extra stock is ordered in anticipation of demand spikes. Participant A said the following: *“At New Year, or Easter, we order extra because we know it’ll sell.”* and Participant E said the similar *“At Christmas, for example, you have to buy a lot of Christmas ham and other items like almonds and such are also bought in greater quantities during holidays.”*. These practices show how MTS is expanded temporarily in seasonal periods, mirroring the dynamic nature of hybrid grocery operations where flexibility in stock strategy is essential.

While MTS makes certain for product availability, it also carries risks of overproduction and food waste. Participant E responds to this with markdown strategies and waste monitoring systems: *“If we see something is nearing its date, we put a special offer/discount on it—like roast beef; if we have three roast beef left, we lower the price and run a special on it.”*. Participant D noted *“We’ve set target figures for everyone who is department responsible and purchasing responsible. We’ve set a goal, how much are we allowed to throw away? We measure both kronor and kilos.”*. These tools are used to mitigate the downsides of MTS when demand does not match expectations. The use of these practical solutions supports Romsdal’s (2014) argument that combining MTS and MTO in hybrid grocery supply chains can support balancing efficiency with responsiveness. MTS provides stability for items with predictable demand, while MTO is more suited for products with short shelf lives or more variable sales patterns. Beckerman (2011) also noted the importance of inventory control to evade capital being tied up in unsold goods, while Nakandala (2019) mentions that hybrid models should adapt to product characteristics to

be effective. These findings illustrate how Swedish grocery retailers use a united inventory strategy to meet service level expectations while managing cost and sustainability pressures.

6. Discussion

The sixth chapter presented an in-depth discussion of the data throughout the thesis chapter. This chapter aimed to compare the qualitative findings together with the theoretical frameworks used. The goal of chapter six was to generate insight and develop new understandings based on the participants' responses and the literature review.

6.1 Research Question & Purpose

This thesis purpose was to get an in-depth understanding of the hybrid supply chains in grocery retail both financially, societal and environmental. The dual pressure from both the cost minimization strategies as well as the sustainability initiatives is significant when finding a balance for the grocery stores to navigate in these complex hybrid supply chain models. The thesis worked to find a balance between the centralized and decentralized models, and to find this balance, trade-offs are required. These trade-offs together with the literature research, empirical findings and analysis contributes to a more deeper understanding.

The main and central question in this thesis is how cost minimization can simultaneously happen while influencing the environment in a hybrid supply chain model and if they can co-exist.

To help investigate this topic the following research question was stated:

How do hybrid supply chain models influence cost drivers, environmental sustainability and social responsibility in the Swedish grocery retail sector?

6.2 Categories connecting to Themes and TBL Framework

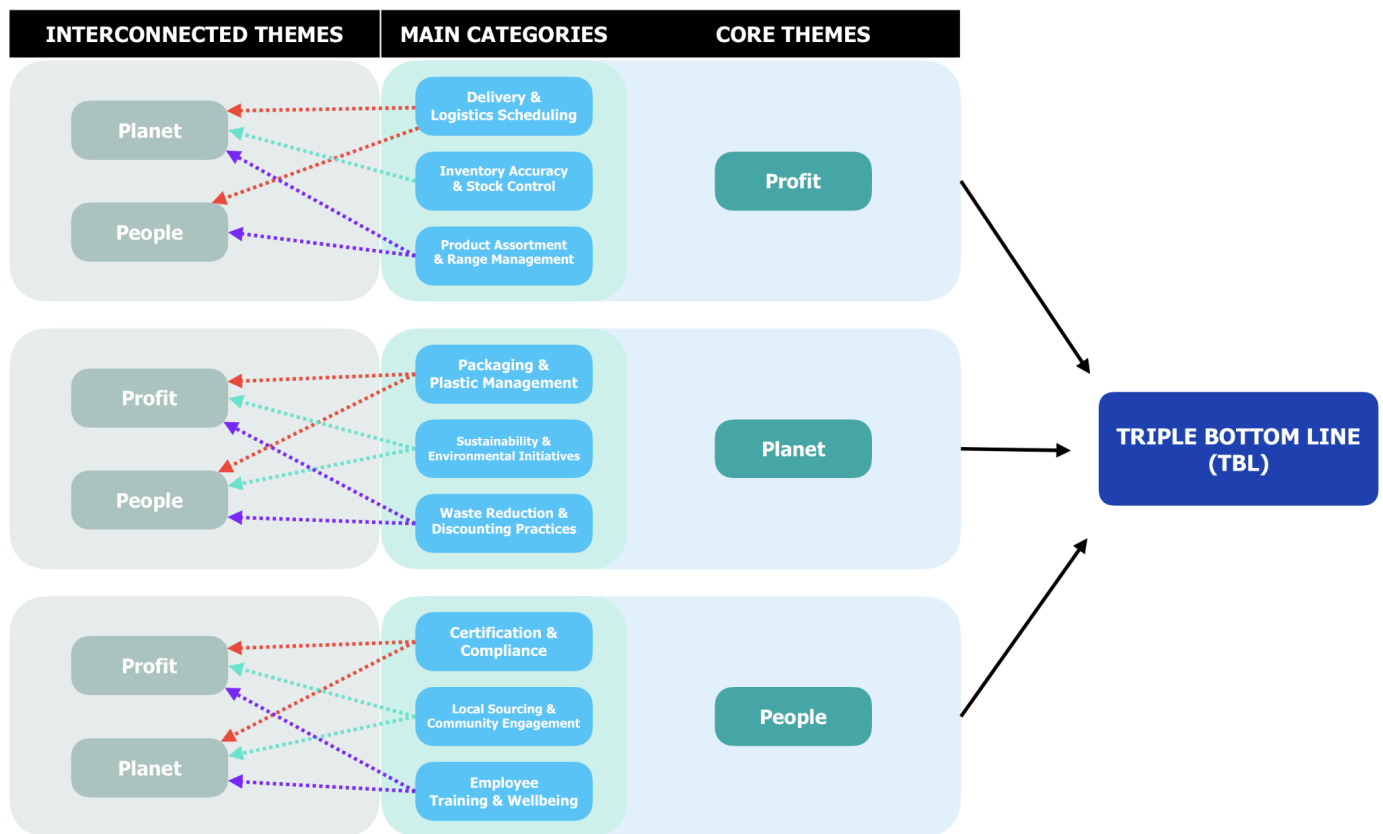


Figure 1

Source: Sedwall, Grandin & Bjerne (2025)

Analysing the thematic data gathered from each participant and connecting each category to the assigned themes, a clear pattern emerged, an alignment with the TBL framework. The themes created, economic, environmental, and social, which can be directly connected to each of the P's, Profit, Planet, and People. Each category specifically focuses on creating connections to the TBL, the one which gained the most valuable insight and different perspectives from the majority of each participant.

6.2.1 Economic (Profit)

The economic theme was the most frequently one brought up throughout the interviews. The three most insightful categories here were delivery and logistics scheduling, inventory accuracy and stock control, and product assortment and range management. The participants explained how these areas optimize the outcomes for profit.

Delivery and logistics scheduling came to be an important factor in managing costs. Retailers strategically use centralized supply chains with regional delivery networks to balance cost efficiency and local market demands. This allows retailers to optimize transportation costs, which Holmberg & Örne (2013) identified as a major cost driver in Swedish grocery logistics, while maintaining flexibility to respond to market variations. Centralizing main logistics operations while keeping distribution flexibility, Swedish retailers create a structural foundation for their hybrid supply chains, sustaining both standardization of high volume goods and customization for regional products.

Inventory accuracy and stock control differed from manual to automated systems across retailers showing that hybrid supply chains exist along a range of technology integration. Finding a balance between automation and human management impacts both economic results through reduced waste, improved stock utilization, and environmental efforts when ordering accurately and reducing surplus. Each retailer customizes their inventory management approach to their specific needs.

The category, product assortment and range management, separate products between standardized items handled on a central level and specialized products managed more flexibly. This method lets retailers achieve efficiency with high volume products while still being able to offer unique or local items. Instead of choosing between efficiency and responsiveness, the hybrid model uses different strategies for product categories based on their characteristics.

6.2.2 Environmental (Planet)

Under the environmental theme, the three most insightful categories here were packaging and plastic management, sustainability and environmental initiatives, and waste reduction and discounting practices. The participants described how these aspects influence supply chain decisions with a specific focus on environmental responsibility.

Regarding retailers facing pressure between sustainability goals, food safety, product preservation, and consumer preferences, as discussed in section 4.2, this follows under the category, packaging and plastic management. The data reveals that these retailers take calculated actions to improve packaging sustainability while maintaining operational practicality in line with the planet aspect of the TBL.

A more general category, sustainability and environment initiatives, beyond packaging, on infrastructure investments like renewable energy systems and environmental certifications. These types of investments often occur in centralized structures, which challenge the standard perception that decentralization is automatically the more sustainable option. The empirical data shows that in hybrid supply chains, environmental improvements depend on making clear decisions about when to centralize and when to decentralize based on where it creates most of the sustainability value.

Another relevant category, waste reduction and discounting practices, connected to the broader environmental theme. Instead of treating waste as a separate environmental practice, the empirical data point out that retailers implement discount systems and inventory controls as combined parts of economic management. This allows them to achieve environmental benefits and manage economic priorities by implementing waste reduction into core business practices.

6.2.3 Social (People)

Three highly relevant categories implied by the empirical data gathered around the core social theme, corresponding to 'people' within the triple bottom line, are certification and compliance, local sourcing and community engagement, and employee training and wellbeing. These categories of elements target social responsibility throughout the supply chain operations of grocery retailers.

Analyzing the coding, a category revolving around certification and compliance is a suitable connection to the social theme. These frameworks expand accountability past organizational borders to include suppliers and partners, strengthening shared accountability across the supply network. Certification guarantees consistent social standards regardless of a supply chain being centralized or decentralized. It is relevant within hybrid supply chains where different structures still require consistent standards to ensure social performance.

Local sourcing and community engagement as a category specifically addressed commitments to work with local suppliers, encouraging the social measurement of supply chain decentralization. Retailers maintain a connection to local communities even though efficiency tensions might otherwise drive toward centralization. The findings indicate that the focus is not only on operations but also on maintaining community connections and supporting local economic development. Mixing decentralized parts that contribute to community development enables them to maintain social support to operate.

Employee training and wellbeing represents a critical category within the social theme. The empirical data reveals that retailers recognize their internal workforce as foundational to successful hybrid supply chain implementation. In 4.3 and 5.3.3, as previously explained, Participant C focus on having regular pulse measurements and performance conversations indicates high levels of effort in this regard. A hybrid model's success ultimately comes down to how motivated and how well trained the employees are.

The categories presented in Figure 2 also interconnect with additional pillars of the TBL framework, creating relationships that impact outcomes on each other, instead of acting independently. It causes both cooperation and trade-offs that retailers have to navigate. For instance, improving inventory accuracy benefits economic performance through cost reduction, but it increases environmental goals and minimizes excessive waste. Working with local suppliers and engaging with local communities strengthens relationships and shortens

transportation routes. This leads to logistics responsiveness and impacts profits while lowering the environmental footprint and supplying more sustainable and healthier produce (Riemer, 2010).

6.3 Aligning Policies with Practice and Corporate Sustainability Goals

The thematic coding illustrates a consistent operational focus on sustainability, cost efficiency, local sourcing, compliance with regulation and social responsibility. When comparing these empirical findings with the official policies of ICA, Coop and Hemköp, there is a general alignment within what the companies promote in their formal sustainability strategies and what is observed in practice. Although, some practical limitations and inconsistencies can emerge, especially around implementation and the tension between sustainability and operational constraints.

From an environmental aspect, all three companies showcase ambitious sustainability targets in their policies. ICA outlines objectives for instance as achieving 100% recyclable or reusable own-label packaging and reducing food waste with support from process innovation (ICA Gruppen, 2024). Coop addresses circular material flows, climate-smart transportation and risk-based product declarations for biodiversity, climate impact and eutrophication (Coop Report, 2024). Hemköp's desire for similar environmental goals includes waste minimization, energy-efficient operations and the usage of sustainable packaging materials (Hemköp, n.d). These corporate goals strongly align with the thematic codes obtained from the interviews, particularly regarding discounting near-expiry products, managing packaging waste and applying circular logistics practices. Participant reflections on markdown boxes, food donations and the use of reusable systems like Svenska Retursystem mirrors the policy focus on minimizing food waste and improving material efficiency. For instance, ICA's documented co-ownership of reusable crate systems is reflected by Participant C's statement about national reuse and cleaning of pallets. However, an ongoing use of plastic gloves for hygiene in fresh food departments, as several participants address, stands in contrast to formal packaging reduction goals. It suggests a policy-practice gap where food safety requirements and customer preferences override sustainability targets in some certain areas.

In terms of social sustainability, the thematic coding features certifications, compliance mechanisms and employee engagement. These are consistently aligned with the documented

policies of Coop, ICA and Hemköp. ICA's Code of Conduct requires supplier compliance with human rights, prohibits child labor and needs employment safety and equal treatment (ICA Gruppen, 2024). Coop reinforces this by its Code of Conduct for suppliers and systematic use of the World Bank's governance indicators to examine social risk (Coop Report, 2024). The interview participants confirmed that goods lacking appropriate certifications could not be sold and discussed traceability and compliance monitoring. This reinforces that regulation is not only policy, but practiced at the operational level. Internal sustainability practices for example as employee well-being tracking and training programs were discussed by the participants, particularly by Participant C, whose account of pulse measurements and performance conversations exemplifies ICA's internal sustainability culture. These insights reflect the value placed on internal engagements found in Hammer & Pivo's (2016) view of social sustainability. Regarding the economic and operational alignment, the coding revealed hybrid supply chain structures, centralization of terminals and investments in digital forecasting tools, which are elements supported by ICA's and Coop's operational strategies. ICA promotes efficiency with automated, centralized logistics and flexible local sourcing (ICA Gruppen, 2024), while Coop addresses supply chain transparency, climate impact reduction and cross-functional procurement reviews (Coop Report, 2024). The interviews confirmed these strategies with detailed explanation of centralized terminal systems, transshipment hubs and digital inventory tools. Participant C's description of a twelve-terminal national network symbolizes ICA's logistics infrastructure. Cost-efficiency goals through route optimization and merged deliveries, as described by the participants, aligns with Coop's sustainability declarations for minimizing transport-related emissions (Coop Report, 2024). Although not all firms explicitly use the term "hybrid supply chain", the operational patterns explained, cross-docking, centralized automation and local supplier integration are consistent with hybrid frameworks discussed in the literature. These strategies illustrate that companies' formal policies are symbolic but are often translated into daily routines and logistical decisions.

In overall, the company policies of ICA, Coop and Hemköp strongly align with the thematic insights drawn from the interviews. A lot of the core elements of the Triple Bottom Line such as cost efficiency, environmental responsibility and social commitments are present in both policy and practice. However, the interviews also revealed that some gaps remain in implementation, especially where day-to-day operations are affected by consumer demands, legal restrictions or

infrastructure limitations. It suggests that while company policies are effective in setting direction and standards, achieving sustainability goals relies on continuous coordination between strategic planning and frontline execution.

7. Conclusion

Building on the previous chapter, this chapter offered a conclusion aimed to address the research question stated in the beginning, which underlay the foundational role of the thesis. This chapter also reflected on the thesis limitations and outlined purposes for future research.

7.1 Conclusion

To conclude, based on our research, hybrid supply chain models significantly impact the costs and environmental sustainability within the Swedish grocery retails. A centralized supply chain logistics combined with decentralized distribution networks, these models help retailers to reduce transportation costs and emissions, also making sure they remain alert to market needs. This balance allows retailers to function efficiently and meet environmental standards and customer expectations. Digital tools further enhance these benefits by improving inventory management and demand forecasting.

Three key operational patterns were identified. First, retailers build centralized distribution centers supported by regionally connected delivery points. Second, they use different inventory strategies depending on product type, with high-volume products following make-to-stock and products with short shelf life following make-to-order. Thirdly, environmental practices are becoming part of core business operations instead of being treated as separate efforts. The theoretical framework, Triple Bottom Line, was established as highly relevant, with economic regards, cost efficiency, inventory optimization, environmental concerns, waste reduction and sustainable packaging, and social commitments, local sourcing, and fair labor. The data explains that cost efficiency and sustainability are not naturally opposing parties.

7.2 Theoretical Implications

The thesis research question focuses on cost efficiency and sustainability, and these two are often viewed as two different goals in the used frameworks. The existing literature which were used throughout our thesis, traditional supply chain, are focusing on either cost minimization or sustainability initiatives and leading to seeing these two as conflicting theories that cant co-exist. For instance centralized logistics can reduce the operational cost but the decentralized sourcing can instead support the local economies and lead to reduced environmental impact, and this shows that these two can co-exist to reach the most optimal level of a hybrid model.

	Cost	Environmental	Social
Centralized	+	-	-
Decentralized	-	+	+

Figure 2

Source: Sedwall, Grandin & Bjerne (2025)

As seen in this trade-off figure we see how there is a clear either or approach in the traditional supply chain where you should choose between one of these approaches. Instead, our thesis shows a support of a more integrated hybrid supply chain and means that organizations do not always need to choose between those two, but can complement each other. This hybrid approach shows that goals such as cost minimization, social views and sustainability can be achieved together when using a combination of centralized and decentralized to balance the trade-offs to achieve optimal level of performance from all perspectives.

The thesis suggests a theoretical implication of a broader and more flexible viewpoint that does not force organizations to choose between two approaches either to being efficient economically or being environmentally responsible. This hybrid model could in turn help future research and strategies how to adapt to societal, environmental and market changes.

7.3 Societal Implications

As seen in the empirical findings, a growing consumer awareness about sustainability is placed on the transparent environmental practices which includes increased interest in local sourcing, reduction of plastic, waste reduction and sustainable logistics in their supply chain strategy. These implications reflect on the strategies required to align their business with consumers purchasing decisions, as seen in the interviews how consumers today demand locally sourced products, and less plastic. The Swedish grocery stores are responding to these consumer demands by focusing on the efficiency and cost minimizations with the local and environmental responsiveness. One focus we saw was the local sourcing practices where all the interviewees had products in their collection sourced nearby.

In addition to the local sourcing awareness, consumers also push for better working conditions and fair work. According to Miemczyk & Luzzini (2019) these strict ethical standards that focus on fair working treatment aligns with the demands from consumers for the corporate social responsibility. By enforcing, and also reporting that the standards are followed like the supplier code of conduct, grocery stores make sure that their organization aligns with the social sustainability goals and consumer awareness. These standards regarding social sustainability according to Hammer & Pivo (2016) need to be monitored and presented to foster a stronger trust against the consumers.

Improved management of sustainability that meets the demands alongside cost minimization would result in broader positive effects. Organizations need to involve the consumers' demands at the same pace as new regulations develop, which also develops the society's viewpoints on the sustainability initiatives. This thesis highlights the increasing importance of societal demands and consumer influence in how the organizations will shape its strategies, since proven in the interviews that aligning strategies with consumer demands increases their overall performance.

7.4 Limitations & Future Research

Even though this study delivers valuable insights into how hybrid supply chain models affect cost drivers, environmental sustainability and social aspects in Swedish grocery retail, various limitations should be acknowledged. Firstly, the amount of interviewees, while sufficient for thematic depth in a qualitative design, limits how widely the findings can be applied. Although diverse roles were included like floor staff and executive leaders, the lack of input from senior

corporate decision-makers or third-party logistics providers could limit insights into higher-level strategic planning and system wide coordination.

Secondly, the study focused heavily on verbal accounts through interviews and company policy documents. There were no direct observations or shadowing of supply chain practices, which may have revealed differences within stated practices and actual behavior. For example, claims about automation usage or waste tracking could be better verified through real-time observation or internal performance data. Including observational or ethnographic methods in further research could acknowledge this gap.

Third, while the study split the data with policy documents from ICA, Coop and Hemköp, it did not involve insights from customers, suppliers or regulators. If these perspectives were integrated in future research, it could help discover more nuanced tensions between sustainability targets, cost constraints and consumer behavior, especially since many participants cited customer expectations as barriers to sustainability (e.g., plastic packaging).

Although the study analyzed sustainability through the Triple Bottom Line lens, it stressed environmental and economic dimensions more heavily than social sustainability. Future research should aim to dig even deeper into employee well-being, labor practices across the supply chain and diversity or inclusion efforts in the sector.

Lastly, the future studies can expand into large-scale analysis to assess how sustainability strategies in hybrid models evolve over time, particularly as regulations tighten and consumer expectations shift frequently. Quantitative methods, such as survey-based benchmarking or cost-impact analysis may complement qualitative findings and include more scalable insights. In conclusion, while this thesis lays an important foundation, more multi-method, stakeholder-inclusive and extensive research is required to comprehensively understand the evolving role of hybrid supply chains in achieving sustainable and cost-efficient grocery retail in Sweden.

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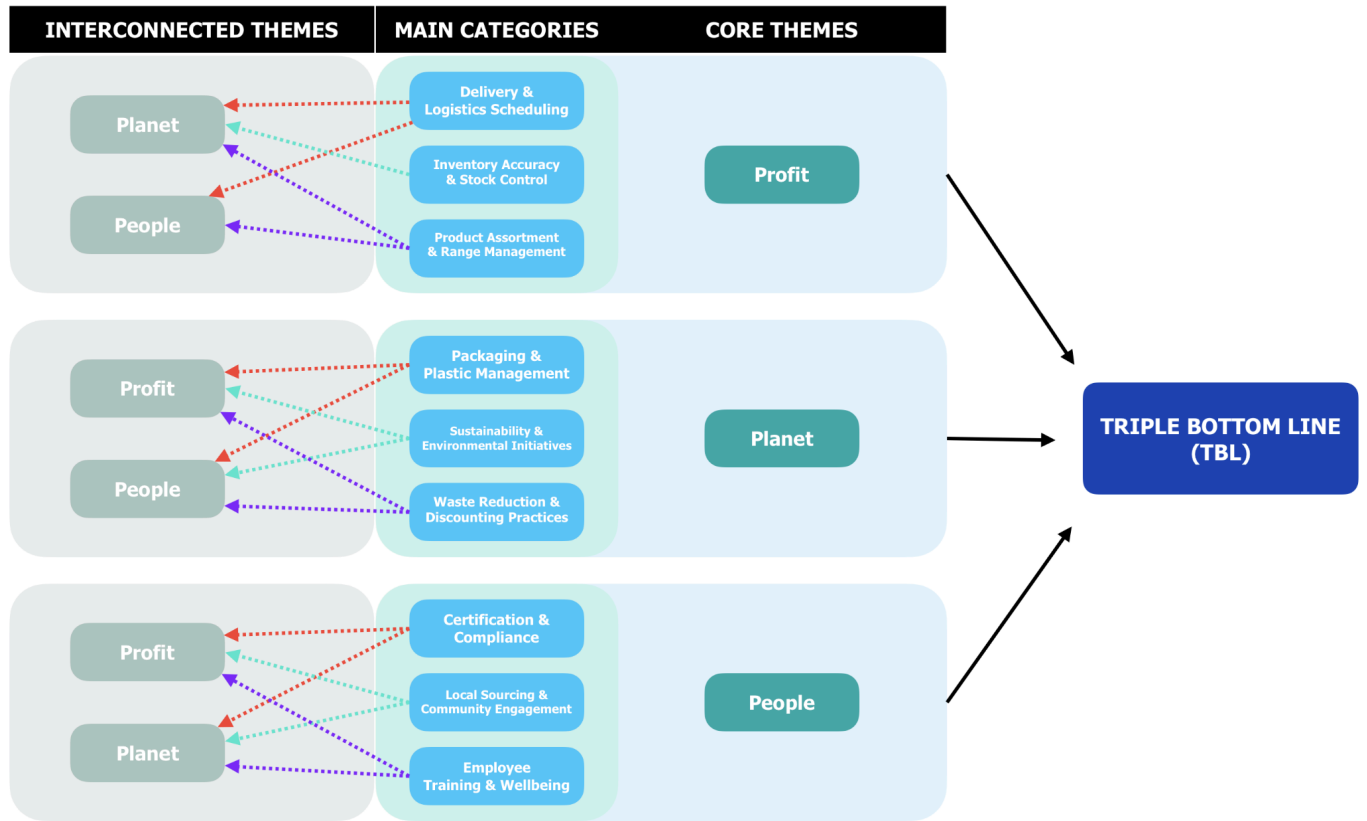
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9. Appendix

9.1 Figure 1 Extended Version of Triple-Bottom-Line



Source: Sedwall, Grandin & Bjerne (2025)

9.2 Figure 2 Centralized vs Decentralized Trade-Offs

	Cost	Environmental	Social
Centralized	+	-	-
Decentralized	-	+	+

Source: Sedwall, Grandin & Bjerne (2025)

9.3 Figure 3 Participant Overview

Participants	Gender	Age Category	Occupation	Date of Interview
Participant A	Female	20-30	Floor-worker	2025-04-02
Participant B	Female	20-30	Floor-worker	2025-04-02
Participant C	Male	50-60	Executive Level	2025-04-03
Participant D	Male	40-50	Executive Level	2025-04-03
Participant E	Female	20-30	Floor-worker	2025-04-23
Participant F	Female	20-30	Floor-worker	2025-04-23
Participant G	Male	30-40	Floor-worker	2025-04-03

Source: Sedwall, Grandin & Bjerne (2025)

9.4 Interview Questions Floor-Workers

(The interviews and questions were conducted in Swedish but is in the appendix translated to English)

1. Can you describe a typical workday for you in the store, focusing on how you receive deliveries and restock items?
2. How is your job affected by customers' purchasing patterns, for example during holidays or trends, when restocking the inventory?
3. Do you feel that you can contribute to the store's work with sustainability and cost-efficiency through your daily tasks?
4. How much influence do you have over when and how much is ordered for the store?
5. How do you think the deliveries work in the store? Do the goods arrive on time?
6. Is your work affected when there are changes in the deliveries?
7. How does the collaboration with suppliers work from your perspective? Do you have any direct contact regarding deliveries or sustainability requirements?
8. How often do problems occur with items being out of stock on the shelves?
9. What times of the year are hard to estimate how much product you need to bring in, and how does this affect waste?
10. Strategies to reduce food waste, what effects have you seen from these?
11. Do you think certain items are frequently thrown away? What could be the reason?
12. How do you usually handle products that customers return or complain about? What do you do to reduce waste in these cases?
13. If you notice something that could be improved in the store's handling of inventory, waste, or sustainability efforts, how do you raise it with your managers?
14. What digital tools or programs do you use to keep track of inventory levels and reduce waste, and how does this affect your work?
15. Do you use any digital tools (tablets, apps) to keep track of the inventory?
16. Are there any initiatives to help vulnerable groups, e.g., through donations or collaborations with charities?
17. Do you have any insight into how the store chooses suppliers based on social sustainability, such as fair working conditions?
18. Do you feel there is a balance between reducing costs and being environmentally conscious in the store?
19. What are your thoughts on the store's efforts to reduce plastic use and packaging materials?
20. Do you receive any training about sustainability or the environment in your stores?
21. Do you notice any difference in your work when you receive organic or eco-labeled products compared to other items?
22. Do you know if the store prioritizes local suppliers over international ones to reduce transport emissions?
23. Do you know where most of the store's products come from and which suppliers we work with?
24. Do you have information about whether your suppliers are certified according to sustainability standards like KRAV, Fairtrade, or MSC?
25. Have you noticed that customers care about sustainability – for example, how products are packaged?
26. Is there anything you would like to change to make your work easier and more sustainable?

9.5 Interview Questions to Executive Leaders

(The interviews and questions were conducted in Swedish but is in the appendix translated to English)

1. Can you describe your role and main responsibilities? How long have you worked here?
2. How do you ensure that staff have the right skills and motivation to achieve your goals related to sustainability and cost control – for example, reducing waste, staying within budget, and working efficiently?
3. Do you mainly use a centralized or decentralized model for inventory and distribution? Why does that model suit your operations best (e.g. central warehouse vs. local suppliers)?
4. How does the cooperation between your central warehouse and local suppliers work?
5. What does your inventory structure look like? Are there specific product categories you always keep in large volumes – and what is the basis for those volume decisions?
6. How do you think deliveries work in the store? Do goods arrive on time?
7. Is your work affected when there are changes in deliveries?
Follow-up: For example, during COVID – how sensitive were you to delivery changes?
8. How do you determine which goods should be ordered as needed (MTO) and which should be kept in stock (MTS)? What influences these decisions – for instance, product type or seasonal demand?
9. How do you balance cost efficiency and environmental responsibility?
10. What do you see as the biggest cost drivers in your supply chain, and how do you work to control them (e.g. transport, warehousing, waste, staffing)?
11. How do you measure and evaluate whether you are meeting your goals in economy environment, and social sustainability?
12. How do consumer demands for sustainability affect your decisions regarding inventory and logistics?
13. Do you have information on whether your suppliers are certified according to sustainability standards like KRAV, Fairtrade, MSC?
14. Do you have any strategies to reduce food waste? What effects have you seen from them?
15. How do you handle products that customers return or complain about? What do you do to minimize waste in these cases?
16. What have you done to make the supply chain more environmentally friendly? Can you give examples of sustainable initiatives in logistics or purchasing (e.g. buying from local suppliers, smarter delivery planning to avoid half-empty trucks)?
17. Have you made any efforts regarding eco-friendly transport or packaging (e.g. protective packaging for storage or sale)?
18. What factors weigh most heavily when you choose suppliers – price, quality, certifications, or environmental responsibility?
19. How do you ensure your suppliers meet your sustainability requirements, and what actions do you take if a supplier fails?
20. What types of goods do you buy most locally, and why those in particular?
21. Are there any digital solutions you currently use to streamline inventory while reducing environmental impact?
22. How do digital tools (like demand forecasting systems) affect your work?
23. How do you manage demand uncertainty to minimize overstock and stockouts?
24. Do you use or consider using logistics solutions like cross-docking (direct transfer from incoming to outgoing transport) to improve supply chain efficiency? Why or why not? (This reduces storage costs, speeds up delivery, and lowers waste – especially for perishables.)
25. What do you see as the biggest future changes in store logistics and purchasing?
26. Do you collaborate with any external actors, such as municipalities or organizations, to improve sustainability in your operations?
27. Do you communicate your sustainability and cost-saving efforts to customers? If so, do you think it affects customer loyalty or sales?
28. Do your goods come from Europe?
29. How do you ensure good working conditions and ethical practices in your supply chain, in line with the Triple Bottom Line (TBL) model?

9.6 Thematic Coding

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9.7 DECLARATION

DECLARATION

I/We hereby declare that this thesis represents my/our own work.

I/We declare that we have obtained the necessary ethical approval (e.g., GDPR) and acknowledged my/our obligations and the rights of the participants/respondents in the research.

I/We have read and applied the Jönköping University's current research ethics guidelines with regards to the use of artificial intelligence (AI) tool in this work as presented in the Course Guide. During the preparation of this work the author(s) used the following AI tools and the purpose for their use.

- AI was used to transcribe, translate, correct grammar, and to find better and more suitable academic words.
- ChatGPT was used to rephrase sentences and improve for a more academic and professional written text.
- Grammarly was used to find more academic wording, correct spelling and punctuation.

As author/s, I/we have reviewed and edited the content as needed and take full responsibility for the content of the thesis.



Signature: Emilia Sedwall

Date: 24/05-2025



Signature: Alexander Grandin

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